

The rational Hodge conjecture for products of K3 surfaces

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Abstract

We prove the rational Hodge conjecture for every finite product of projective complex K3 surfaces: every rational Hodge class is algebraic. The factors may be distinct or repeated, with no restrictions on their Picard numbers, periods, or endomorphism fields.

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1 Introduction

For a smooth projective complex variety X , an algebraic subvariety of codimension p has a rational cohomology class of Hodge type (p, p) . The rational Hodge conjecture asks whether these classes span every rational class of that type. Write

$$\mathrm{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X, \mathbb{C}),$$

where the intersection is taken inside $H^{2p}(X, \mathbb{C})$, and write $\mathrm{CH}^p(X)_{\mathbb{Q}}$ for codimension- p cycles modulo rational equivalence with rational coefficients.

Theorem 1.1. *Let $m \geq 1$, and let $S_1, \dots, S_m$ be any projective complex K3 surfaces. For $X = S_1 \times \dots \times S_m$ and every $0 \leq p \leq 2m$, the cycle-class map*

$$\mathrm{cl} : \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X)$$

is surjective.

The factors may repeat, and no relation between their periods or endomorphism fields is assumed. The distinction matters: a product can contain Hodge tensors coupling different transcendental spaces, while a power of one surface can contain tensors beyond those generated by pairwise contractions. The proof treats both phenomena.

For a projective K3 surface S , its transcendental space is

$$T(S) = \mathrm{NS}(S)_{\mathbb{Q}}^{\perp} \subset H^2(S, \mathbb{Q}),$$

with the intersection form.

1.1 History and the remaining tensor problem

The Hodge conjecture, formulated in Hodge’s 1950 Congress address [Hod52], asks which rational cohomology classes come from algebraic cycles. For a projective K3 surface itself, the Lefschetz $(1, 1)$ theorem, together with the classes of the surface and of a point, answers the question. Products introduce a different problem: their Hodge classes include tensors in the transcendental cohomologies of the factors. Already for two factors, the nontrivial part concerns Hodge homomorphisms between these cohomologies; on higher powers, further invariant tensors occur. We use the customary background on K3 surfaces, their periods, and their transcendental Hodge structures as presented in [Huy16].

Kuga and Satake associated an abelian variety to a polarized Hodge structure of K3 type [KS67]. Its first cohomology carries a Clifford-algebra representation, and the original weight-two Hodge structure embeds into a tensor square of this weight-one structure. Deligne developed this construction in families and used it in his proof of the Weil conjecture for K3 surfaces [Del72, Sections 5–6]. André subsequently proved that the Kuga–Satake correspondence is motivated [And96, Section 7.1]. These results provide the Hodge-theoretic and motivic relation to abelian varieties; obtaining an algebraic cycle representing the correspondence is the additional requirement in the Kuga–Satake Hodge conjecture. Van Geemen’s account explains the Clifford construction and its relation to exceptional Hodge classes on abelian varieties [vG00].

A second line of work constructs correspondences between K3 surfaces using moduli spaces of sheaves. Mukai’s work supplied the fundamental examples [Muk87], and Nikulin extended the isometry result using quadratic-lattice methods [Nik88]. Buskin proved that every rational Hodge isometry between the second cohomologies of two projective K3 surfaces is algebraic [Bus19, Theorem 1.1]; Huybrechts gave a proof using twisted derived categories and established an isomorphism of their Chow motives [Huy19, Theorems 0.1–0.2]. The isometry hypothesis is significant: arbitrary Hodge homomorphisms need not preserve the quadratic forms. Varesco proved that algebraic Kuga–Satake correspondences also make Hodge similarities algebraic [Var23, Theorem 4.5].

Zarhin determined the Hodge groups of transcendental K3 cohomology in terms of its endomorphism field, which is either totally real or CM [Zar83]. This description explains why a result about squares need not settle the problem for powers. In the totally real case, orthogonal invariant theory produces determinant tensors in addition to the quadratic contractions. Varesco identified the corresponding relative determinant classes and proved a precise generation statement [Var25, Theorem 2.16]. When the endomorphism field is CM, the invariants are generated in degree two; the work of Ramón Marí, together with algebraicity of Hodge isometries, gives the Hodge conjecture for all powers [RM08, Bus19]; see also [Var25, Theorem 2.2 and Corollary 2.3]. Here a CM endomorphism field is the relevant condition; it need not make the whole transcendental Hodge structure of CM type.

Algebraic Kuga–Satake correspondences have also been constructed for important special families. Paranjape treated the K3 surfaces arising from double covers of the plane branched along six lines

[Par88], and Schlickewei used their Kuga–Satake varieties to establish the Hodge conjecture for their squares [Sch10, Theorem 2]. Varesco proved an all-powers theorem for four-dimensional families of generic Picard number 16 with a specified rational transcendental quadratic space, assuming algebraicity of the Kuga–Satake correspondences throughout the family [Var25, Theorem 0.2]. The quadratic-space and family hypotheses in that statement are essential to its scope.

Recent geometric constructions considerably enlarge these cases. Floccari proved algebraicity of the Kuga–Satake correspondence and the Hodge conjecture for all powers of the K3 surfaces associated with varieties of generalized Kummer type [Flo26, Theorem 1.2]. Floccari and Fu obtained related results through singular varieties of O’Grady’s six-dimensional deformation type; their construction also gives products of powers of the abelian, K3, and hyper-Kähler varieties associated with one such singular variety [FF26, Theorem 5.7 and Corollary 5.8]. These theorems illustrate how geometric relations with abelian varieties control exceptional tensors on particular families. For a general list of K3 surfaces, a further issue is to control the joint Hodge group: algebraicity of the tensors on each individual factor does not by itself determine the mixed invariants.

1.2 The proof and its principal constructions

The Kuga–Satake realization of $T(S)$ uses the weight-one Hodge structure on the full even Clifford algebra $W_S = C^+(T(S))$, the H^1 of an abelian variety up to isogeny. For a rational vector $w \in T(S)$ of nonzero square, the standard correspondence is the specified Hodge map

$$T(S)(1) \longrightarrow \text{End}(W_S), \quad v \longmapsto (x \longmapsto vxw). \quad (1.1)$$

Here $\mathbb{Q}(1)$ has weight -2 . A polarization converts this into a degree-two tensor on the square of the abelian variety. The precise map in (1.1) is essential to the argument.

The bridge from geometry to this exact tensor is Proposition 2.1. It starts with a rational middle-dimensional class killed by the symplectic form on a quartic times a large standard torus, or on the torus alone. Under a condition on its degree-two annihilator, it produces a perfect complex on the ordinary mirror. The conclusion controls two different features of the complex: its Euler pairings with specified test objects and the dimension of its second self-extension space. Part I first uses this proposition to prove Theorem 1.1; Part II then proves the proposition.

The force of the realization statement lies in combining its two conclusions. In the K3 construction, the Euler pairings initially determine only the algebraic component of the complex’s Mukai vector. We choose that component using Clifford multiplication and separated powers of invariant alternating tensors. The Hochschild action on the whole vector factors through Ext^2 , so the small extension space forces many cohomological operators to annihilate it. A Clifford commutant calculation shows that so large a kernel is possible only for the prescribed transcendental component, with its exact normalization. Thus an upper bound on self-extensions determines a tensor that the Euler pairings cannot see.

The same calculation attains equality in the rank bound. Equality makes the semiregularity trace injective, so the complex formally deforms along its Hodge locus. This uses the deformation principle developed by Buchweitz–Flenner and Pridham [BF03, Pri24]; Markman’s construction of Weil classes by deforming semiregular sheaves gives a related application [Mar25, Sections 7–9]. We apply this principle along the coupled K3 and Kuga–Satake period family. Algebraization and specialization give (1.1) on every projective K3 surface, as proved in Theorem 4.1.

This use of an extension-space bound is useful beyond identifying one tensor: it obtains the required Hochschild rank on the algebraic mirror itself. It requires no identification of the Hochschild action under the mirror equivalence. The same method, applied to a torus alone, constructs the degree-four spin tensors of Proposition 5.1.

Two companion results enter the final algebraic argument. The exact full-even-Clifford criterion of [Ope26b, Theorem 1.1] gives all Hodge classes on self-powers once (1.1) is algebraic. The rational Hodge theorem for CM abelian varieties [Ope26d, Theorem 1.1] controls the torus constituents of auxiliary weight-one Hodge structures. These are theorem inputs with the stated scope; the proof below supplies the exact KS correspondences and the additional tensors needed to combine different surfaces.

For the combination, compare the group preserving algebraic correspondences with the joint Hodge group of the transcendental spaces. The self-power results identify the projections of these groups to each factor; the remaining question is whether the relations between different factors agree. Algebraic morphisms between weight-one abelian realizations identify the shared simple factors projectively. Central signs can still differ, especially among the low-rank orthogonal and unitary factors sharing $\mathfrak{sl}_2$. The degree-four spin tensors give perfect pairings between local Hom spaces and transport those signs across a finite grid of representation choices. A graph calculation then shows that the remaining signs come from the center of the joint derived group. This is the mixed-product step in Section 6; self-power algebraicity alone does not replace it.

It remains to prove the realization proposition. We represent the input class, after multiplication, by a spin graded Lagrangian immersion whose second cohomology is controlled by the class's degree-two annihilator. Its self-intersection generators have odd integer degrees. A disk-moduli Stokes identity makes the immersed object unobstructed, and its degree-two endomorphism space is then bounded by that second cohomology. Only the embedded test objects are passed through ordinary mirror symmetry. A directed comparison and a categorical projection produce the perfect complex while retaining the Euler pairings and the extension bound. The geometry adapts the controlled-immersion construction of [Ope26c]. Sections 7–10 provide the additional arguments for a K3 factor, arbitrary large torus dimension, integer gradings, the Stokes identity, and the categorical comparisons.

1.3 Conventions and organization

All cycle classes and correspondences have rational coefficients. An algebraic tensor means a cohomology tensor represented by a rational algebraic correspondence; no assertion about relations modulo rational equivalence is implicit. Hodge groups use the special, or circle, convention and fix Tate structures. In calculations with total cohomology, Tate twists equalizing weights are left implicit only when the circle action makes their meaning unambiguous.

The symplectic constructions use spin branes, zero relative-spin background, and integer Maslov gradings. Novikov fields and their extensions are embedded in $\mathbb{C}$ as abstract fields; these embeddings are not analytic evaluation of a Novikov parameter. The relevant mirror objects are fixed before the rational input class is chosen.

In Part I, Section 2 states the realization proposition and analyzes its mirror tests. Section 3 develops cap-action and deformation tools. Sections 4 and 5 construct algebraic spin tensors; Section 6 uses them to prove Theorem 1.1. Part II proves the proposition: Sections 7 and 8 construct the immersion and remove its curvature; Sections 9 and 10 compare the categories.

Part I

Algebraic tensors and the Hodge conjecture

2 A realization statement and its mirror test objects

The first step is to realize a rational middle-dimensional class by a perfect complex whose second self-extension group is small. The realization will preserve its pairings with a fixed collection of Lagrangian test objects. We specify those objects here and then use the realization statement to construct the algebraic tensors. Its proof occupies Sections 7–10 in the second part of the paper.

2.1 The symplectic spaces and the realization statement

Let Y be either a smooth quartic K3 surface with the integral symplectic form ω_Y representing its hyperplane class, or a point. In the latter case all Y -factors below are omitted. Set

$$T_g = \mathbb{R}^{2g} / \mathbb{Z}^{2g}, \quad \omega_T = \sum_{j=1}^g q_j \wedge p_j, \quad D = Y \times T_g, \quad 2n = \dim_{\mathbb{R}} D. \quad (2.1)$$

Here q_j, p_j denote the coordinate one-forms; we use the same letters for coordinates when describing graphs. Write ω for the product symplectic form. Throughout this construction g is even and $n \geq 12$. All branes have integer Maslov gradings, spin structures with zero background class, and the orientations determined by their gradings.

The ordinary mirrors are the mirror quartic B of Y and the product E^g , where E is the Tate elliptic curve; set B equal to a point when Y is a point. These varieties are initially defined over Novikov fields. The change of parameter in quartic homological mirror symmetry is included in the definition of B . We allow further field extensions and then an abstract field embedding into $\mathbb{C}$. Thus passage to a complex variety means extension of scalars along a field embedding, not evaluation of a Novikov series at a complex number. All fields used below have cardinality at most the continuum.

Choose a finite collection of graded Lagrangian spheres $\Sigma_\ell \subset Y$ which split-generates the quartic Fukaya category and whose classes span

$$P_Y = \omega_Y^\perp \subset H^2(Y, \mathbb{Q}). \quad (2.2)$$

Their existence is recalled below. For the point use one test object, with the unit convention. In T_g choose finitely many graphs

$$G_M = \{p = Mq\}, \quad M \in \text{Sym}_g(\mathbb{Z}), \quad (2.3)$$

and make small Hamiltonian perturbations when transversality is required. The graph gradings vary continuously with the symmetric graph planes; the spin structures come from the base frame. The distinct slopes are totally ordered by positive-definite difference. We include a generating set of scalar slopes and may enlarge the list as described below. This order on slopes is separate from any directed ordering of their Hamiltonian copies.

Denote the mirror images of Σ_ℓ and G_M by Q_ℓ and N_M . The graph images are line bundles after one common shift. Put $K_{\ell, M} = \Sigma_\ell \times G_M$. For perfect complexes we use the Euler pairing $\chi(A, F) = \sum_j (-1)^j \dim \text{Ext}^j(A, F)$.

Proposition 2.1 (Realization with a self-extension bound). *Fix the spaces and test systems above. Let $a \in H^n(D, \mathbb{Q})$ satisfy $\omega \smile a = 0$, and define*

$$\mathcal{Z} = \{x \in H^2(D, \mathbb{Q}) : x \smile a = 0\}.$$

Suppose that cup product induces an injective map

$$H^1(D, \mathbb{Q}) \otimes \mathcal{Z} \longrightarrow H^3(D, \mathbb{Q}). \quad (2.4)$$

After a field extension and an abstract embedding into $\mathbb{C}$, there is $\mathcal{E} \in \text{Perf}(B \times E^g)$ such that

$$\dim \text{Ext}^2(\mathcal{E}, \mathcal{E}) \leq b_2(D) - \dim \mathcal{Z}. \quad (2.5)$$

There is a common nonzero $\lambda \in \mathbb{Q}$, independent of ℓ, M , with

$$\chi(Q_\ell \boxtimes N_M, \mathcal{E}) = \lambda \langle a, [K_{\ell, M}] \rangle. \quad (2.6)$$

The mirror varieties, their enhanced categories, and the images of the fixed test systems can be chosen before a is specified.

The last assertion lets us choose divisor classes and ample classes on the geometric mirror before choosing a . One may use a preliminary complex embedding to describe their lattices and cones: the geometric Néron–Severi lattice, intersection pairing, and ampleness of its classes are unchanged by algebraically closed field extension. Later embeddings therefore preserve the algebraic data used in the construction.

The proof of Proposition 2.1 first constructs a graded Lagrangian immersion representing a multiple of a . Its degree-two cohomology gives the right-hand side of (2.5). Unobstructedness places it in a proper cyclic category. We then identify the subcategory generated by the embedded test objects with $\text{Perf}(B \times E^g)$ and project to that subcategory. The categorical completion is given in Section 10.

2.2 Sphere tests and the algebraic Mukai space

In this subsection assume that Y is a quartic K3 surface. Seidel’s quartic theorem gives the enhanced equivalence on a generating category and split-generation by vanishing spheres [Sei15, Theorem 1.3, Corollary 9.6, and Lemma 11.4]. Rescaling the Novikov parameter accommodates the normalization of ω_Y . Sphere branes are rational: no one-cycle holonomy condition has to be imposed. Compatibly graded isotopic spheres determine quasi-isomorphic objects by [Sei15, Lemma 8.14]. These facts apply in particular to all the Hamiltonian copies used later.

We may add vanishing spheres from a generic pencil of quartics in $\mathbf{P}^3$. Symplectic transport identifies its smooth fibers with Y , and the vanishing cycles span the kernel of ambient inclusion. Their Poincaré duals therefore span P_Y . Equivalently one can use a hyperplane pencil in the corresponding Veronese embedding.

For any finite or countable set of the embedded sphere copies, there is a common compatible almost complex structure with no nonconstant holomorphic spheres and no nonconstant holomorphic disks on any individual copy. For somewhere injective curves the unmarked expected dimensions are respectively -2 and -1 . Genericity, sphere factorization, and the disk decomposition results of Kwon–Oh and Lazzarini [KO00, Laz00] give the assertion, as used in [Sei15, Section 8]. The torus graphs have vanishing relative second homotopy, so a product almost complex structure has the same exclusion property for the embedded product tests.

Write

$$P_B = H^0(B, \mathbb{Q}) \oplus \text{NS}(B)_{\mathbb{Q}} \oplus H^4(B, \mathbb{Q}), \quad (2.7)$$

with the Mukai pairing, in which H^0 and H^4 pair with a minus sign. Before choosing a complex embedding, this denotes the geometric divisor space together with its degree-zero and degree-four generators. For an object Q its Mukai vector is $v(Q) = \text{ch}(Q)\sqrt{\text{td}(B)}$.

Lemma 2.2. *The space P_B has dimension 21. Sending the Poincaré dual of a sphere test to the Mukai vector of its mirror image induces a rational isometry*

$$\rho : P_Y \xrightarrow{\sim} P_B. \quad (2.8)$$

Proof. The Euler pairing of two sphere branes is minus their intersection pairing. On their mirror objects it is minus the Mukai pairing. Since the test classes span the nondegenerate, 21-dimensional space P_Y , their Mukai vectors span a space of dimension at least 21.

The mirror is the geometric generic member of the one-parameter family obtained by resolving the diagonal-group quotient in Seidel's theorem. That family has nonconstant period. Before taking the finite symplectic quotient, this is visible in the quartic pencil generated by the product of the four coordinates and their fourth powers: at the Fermat member the infinitesimal period in the product direction is nonzero, because $x_0x_1x_2x_3$ survives in the Jacobian ring. The group fixes the holomorphic two-form. Pullback by the quotient correspondence over an open base therefore transports this variation to the resolved family. Its geometric generic Picard rank is at most 19 by period theory. The HMS parameter is transcendental, including after its formal reparametrization, and geometric Picard rank is unchanged by algebraically closed field extension. Hence $\dim P_B \leq 21$.

We have equality. The pairing equality now also proves that the map on sphere classes is well-defined: a relation among their classes maps to a vector orthogonal to a spanning set of the nondegenerate space P_B , and thus to zero. The resulting surjective map preserves the pairings, which proves (2.8). $\square$

2.3 Graph images and full faithfulness on the test systems

We use Abouzaid's family Floer theorem in its finite-object form [Abo17, Theorem 2.10 and Appendix A]. It gives a faithful A_{∞} functor to analytic perfect complexes. For the present test systems a dimension comparison will give full faithfulness.

The fibration $T_g \rightarrow \mathbb{R}^g/\mathbb{Z}^g$ is nonsingular, and both its base and total space have vanishing second homotopy. A graph induces $b \mapsto (b, Mb)$ on fundamental groups, so $\pi_2(T_g, G_M) = 0$. The graphs and their Hamiltonian copies are therefore tautologically unobstructed. The global zero section and the base frame trivialize the section and spin twists in family Floer theory. Let $\mathfrak{q}$ be the Tate parameter, of positive valuation. The rigid analytic mirror has multiplicative coordinates $z = (z_1, \dots, z_g)$ and periods $z \mapsto \mathfrak{q}^b z$, and is $(E^g)^{\text{an}}$.

Lemma 2.3 (Graph images). *For each finite test system, the image of G_M is a line bundle up to shift, with numerical type εM for one common sign $\varepsilon \in \{1, -1\}$. After fixing the variance convention, all shifts can be made equal. The family Floer functor is fully faithful on the finite systems and their Hamiltonian copies.*

Proof. A graph has one transverse intersection with each torus fiber. Its local family Floer complex is consequently a rank-one module in one degree. We recall the numerical calculation of [Ope26c, Lemma 9.1], including its extension to non-diagonal slopes. On the real cover of the base lift a perturbed graph as $p = Mq + \nabla f(q)$, where f is periodic. Comparison paths along this lift have

zero character offset: the fixed lattice offset of a lifted graph boundary cancels at its two ends, and a residual fiber loop could not bound. Scalar trivializations on the real cover therefore give the period multiplier

$$j_M(b, z) = c_b z^{-\varepsilon M b}, \quad b \in \mathbb{Z}^g, \quad c_b \in \Lambda^*. \quad (2.9)$$

Here Λ is the Novikov coefficient field and a change of variance by perfect duality changes all numerical signs together.

Every integral symmetric matrix is a sum of signed matrices uu^t , with $u \in \mathbb{Z}^g$. Products of theta lines pulled back under $E^g \rightarrow E$, $(x_j) \mapsto \sum u_j x_j$, have precisely the prescribed type εM . Dividing their multiplier into (2.9) leaves a scalar character, hence a line in $\text{Pic}^0(E^g)$. Projective rigid GAGA algebraizes the image. For the enhanced comparison one can use the common-refinement Čech models of [Ope26c, Section 9]; their restriction and cup maps give the same dg comparison in g variables.

For distinct slopes with definite difference, Floer cohomology has dimension $|\det(M - M')|$ and is concentrated in one degree. The quotient of the corresponding lines is ample or anti-ample, and its total Ext dimension is the same determinant. Faithfulness is thus fullness for these pairs. For copies of the same slope, self-Floer cohomology has the exterior-algebra dimensions of a g -torus. A nontrivial Pic^0 quotient would have zero cohomology, contradicting faithfulness. The underlying lines of the copies therefore agree, and their self-Ext dimensions again give fullness.

Choose the arrow direction in which definite graph comparisons have degree zero. The numerical sign must make the corresponding line quotients ample. Otherwise every pair among three ordered scalar slopes would require the same nonzero shift difference g , which contradicts additivity of shift differences. With the correct sign, all scalar shifts agree. Comparison with extreme scalar slopes makes the shift of every other graph the same. Grading transport and the self-cohomology support $[0, g]$ give the same conclusion for copies. This is the normalization argument of [Ope26c, Lemma 9.2]. $\square$

2.4 Finite generation and the order of choices

Lemma 2.4. *Let H be the principal product theta line on E^g . There are integers $k_1 < \dots < k_{g+1}$ such that, for every choice $P_j \in \text{Pic}^0(E^g)$, the lines $H^{-k_j} \otimes P_j$ split-generate $\text{Perf}(E^g)$. The assertion persists under algebraically closed field extension.*

Proof. Choose a vector-bundle generator V of $\text{Perf}(E^g)$ [BV03]. Starting with $K_0 = V$, take evaluation surjections

$$0 \longrightarrow K_j \longrightarrow W_j \otimes (H^{-k_j} \otimes P_j) \longrightarrow K_{j-1} \longrightarrow 0.$$

Here W_j is a finite-dimensional vector space. The exponents can be chosen successively and uniformly in all preceding P_j : perform each step over the proper parameter space $(\text{Pic}^0 E^g)^j$, using its Poincaré lines and relative Serre generation and vanishing. The kernels remain vector bundles. The length- $g + 1$ sequence gives a triangle

$$K_{g+1}[g] \longrightarrow C_V \longrightarrow V \longrightarrow K_{g+1}[g + 1],$$

where C_V is a complex of sums of the displayed lines. Since $\text{Ext}^{g+1}(V, K_{g+1}) = 0$, the triangle splits and V is a summand of C_V . All evaluation sequences and base-change properties were constructed over the parameter spaces, so the argument persists after field extension. The original generator remains a generator after extension, for example by the usual perfect-diagonal criterion. This is the argument of [Ope26c, Lemma 9.3] in dimension g . $\square$

If the sign convention gives ample rather than anti-ample powers, dualize the preceding construction. Thus the scalar graph tests can always be chosen to include generators. Further tests may be added while retaining definite order: the integral symmetric matrices larger than a fixed matrix in positive-definite order are Zariski dense. Consequently every finite-dimensional span obtained by polynomial evaluation on symmetric matrices can be obtained by continuing the ordered list. Extreme scalar slopes can then be added for the shift normalization.

We fix these finite systems and their mirror images before choosing a . Generation in both factors survives the later field extensions. Finally, E is non-CM after every complex embedding used here. Its j -invariant has negative Novikov valuation, so is transcendental over $\mathbb{Q}$, whereas a CM elliptic curve has an algebraic j -invariant.

3 Cohomological operators and deformation

Proposition 2.1 specifies Euler pairings and bounds an Ext^2 space. We now express both conclusions in linear algebra. The Euler pairings determine an algebraic component of a Mukai vector; the bound constrains the kernel of the Hochschild cap action on the entire vector. When that kernel has the expected dimension, the semiregularity map is injective, and the complex formally deforms along its Hodge locus. Spreading and specialization then transport the algebraicity of its Chern character. This is the mechanism used in Sections 4 and 5.

3.1 The linear dictionary

Two gradings will enter the argument. To choose the symplectic input class, we use the Lefschetz action on the cohomology of D . To study the resulting complex, we use the actual Hodge grading on the mirror. We first describe the former as a linear model for the latter; identifying their transcendental Hodge structures will require the later rank argument.

We use the Mukai form q on the total even cohomology of a K3 surface: it is the intersection form on H^2 , and $q(1, [\text{pt}]) = -1$. Mukai vectors are $v(\mathcal{E}) = \text{ch}(\mathcal{E})\sqrt{\text{td}}$. When using the Hodge circle, we suppress Tate twists; invariance means type (p, p) separately in each cohomological degree.

Suppose first that Y is a quartic. Recall the isometry $\rho : P_Y \rightarrow P_B$ supplied by the sphere tests in Section 2, where

$$P_Y = \omega_Y^\perp \subset H^2(Y, \mathbb{Q}), \quad P_B = H^0(B, \mathbb{Q}) \oplus \text{NS}(B)_\mathbb{Q} \oplus H^4(B, \mathbb{Q}).$$

The orthogonal complement of P_Y in the Mukai space is

$$U = \langle 1, \omega_Y, [\text{pt}] \rangle.$$

It is a split ternary quadratic space. Lefschetz theory gives an $\mathfrak{sl}_2$ -action with Cartan operator $2 - \text{deg}$ and lowering operator $f = \omega_Y \cup -$. The space U is its irreducible representation of highest weight two, while P_Y is trivial. Set $t_+ = 1$ and $t_- = -[\text{pt}]$, so $q(t_+, t_-) = 1$, and choose a weight-zero basis vector t_0 of U . We identify $\text{Spin}(U)$ with SL_2 . Write $U_B = T(B)$. Witt cancellation in the two rational Mukai lattices identifies U , as a quadratic space, with U_B . This is a rational quadratic identification only; no assertion about the transcendental action of mirror symmetry is used.

For the torus write

$$W = M_0 \otimes H, \quad \dim M_0 = g, \quad H = \mathbb{Q}e_+ \oplus \mathbb{Q}e_-, \quad f(e_+) = e_-.$$

The vectors e_+, e_- are the standard SL_2 -basis, of weights $1, -1$. Write a_i, b_i for the corresponding basis vectors of W , $1 \leq i \leq g$. The creation–annihilation transform is the linear isomorphism

$$\Phi : H^*(T_g, \mathbb{Q}) \longrightarrow \bigwedge W \quad (3.1)$$

that sends $p_1 \cdots p_g$ to 1, cup product by q_i to exterior multiplication by b_i , and cup product by p_i to contraction by a_i^* . With consistent ordering, the other two creation–annihilation rules are

$$\Phi \iota_{q_i^*} \Phi^{-1} = \iota_{b_i^*}, \quad \Phi \iota_{p_i^*} \Phi^{-1} = a_i \wedge -.$$

Together these four rules and the chosen vacuum characterize the transform. Thus the Cartan operator on the source is $g - \deg$, and cup product by ω_T becomes f . The transform is not a ring isomorphism.

For a symmetric integral slope matrix M , the graph class is, up to a fixed orientation sign, $\prod_i (p_i - (Mq)_i)$. Consequently

$$\Phi(\mathrm{PD}[G_M]) = \pm \exp \left(\sum_{i,j} M_{ij} a_i b_j \right). \quad (3.2)$$

One may verify the identity by the annihilating operators of the graph and its constant coefficient. Mark $H^1(E, \mathbb{Q})$ by H , and hence $H^1(E^g, \mathbb{Q})$ by W . The numerical graph calculation in Section 2 makes the right side the Mukai vector of the mirror line, with the common shift understood. The marking is oriented or reverse oriented according to the single sign convention for all graph lines; it is fixed throughout the construction.

The degree-two SL_2 -invariants in $\bigwedge W$ are precisely the expressions $\sum M_{ij} a_i b_j$ with M symmetric. The first fundamental theorem for SL_2 says that they generate its exterior-algebra invariants [Wey46]. Expanding the exponentials and polarizing therefore shows that the classes in (3.2) span those invariants. A sufficiently large finite graph list has the same span. Since E is non-CM, these are exactly its power’s rational Hodge classes. Their Euler pairing is nondegenerate and agrees, up to one fixed nonzero scalar, with the intersection pairing under Φ : on pairs of graph exponentials this is the determinant formula, and those classes span.

Combine the torus transform with the sphere isometry to define

$$\Psi = (\rho \oplus \mathrm{id}_U) \otimes \Phi : H^*(Y, \mathbb{Q}) \otimes H^*(T_g, \mathbb{Q}) \xrightarrow{\sim} (P_B \oplus U) \otimes \bigwedge W. \quad (3.3)$$

Here the target still contains the formal Lefschetz space U , rather than U_B with its actual Hodge structure. When Y is a point, set $\Psi = \Phi$.

We can now state precisely what the Euler tests recover. In the quartic case let

$$C_* \in (P_B \oplus U) \otimes \bigwedge W \quad (3.4)$$

be rational and invariant under the simultaneous SL_2 , acting trivially on P_B . In the torus-only case take $C_* \in \bigwedge W$ invariant under SL_2 . Set $a = \Psi^{-1}(C_*)$. This lies in $H^n(D, \mathbb{Q})$: total Cartan weight zero means middle degree, and SL_2 -invariance gives $\omega a = 0$. If the annihilator of a satisfies (2.4), Proposition 2.1 gives a perfect complex. Divide its Mukai vector by the common nonzero rational scalar in the resulting Euler comparison, including the fixed pairing normalization of Ψ , and call the result C . Then

$$C|_{P_B} = C_*|_{P_B}; \quad \text{in the torus-only case, } C = C_*. \quad (3.5)$$

Indeed the sphere tests lie in P_Y , and the graph tests span all the relevant torus Hodge classes. The P_B -component of an algebraic class is itself Hodge. The nondegenerate pairings therefore determine that component. They do not yet determine the component in $T(B)$.

3.2 The annihilator and cap operators

The degree-two annihilator of a can be calculated directly on C_* . A K3 degree-two class becomes $x \in P_B \oplus \langle t_0 \rangle$. Its cup action is

$$D_x(t_+) = x, \quad D_x(y) = -q(x, y)t_- \quad (y \in P_B \oplus \langle t_0 \rangle), \quad D_x(t_-) = 0. \quad (3.6)$$

The torus degree-two actions are all operators

$$F(\lambda, L, \pi) = \lambda \wedge - + F_L + \iota_\pi, \quad \lambda \in \bigwedge^2 \langle b_i \rangle, \quad \pi \in \bigwedge^2 \langle a_i^* \rangle, \quad (3.7)$$

where F_L is the derivation sending the a -copy of M_0 to the b -copy by an arbitrary matrix L , and vanishing on the b -copy. On the tensor product the full operator is $D_x \otimes 1 + 1 \otimes F(\lambda, L, \pi)$. These formulas follow from cup product on a surface and the creation–annihilation rules.

The actual Hochschild cap action on C has the same operator formulas, after choosing bases for its Hodge weights. The two calculations have different inputs:

	formal calculation	actual calculation
class	C_*	C
K3 complement	U	$U_B = T(B)$
weights	$n - \deg$ on $H^*(D)$	$p - q$ on $H^{p,q}$
operators	$H^2(D)$ by cup product	$\mathrm{HH}^2(B \times E^g)$ by cap product

Omit the K3 row and the factor B in the torus-only case. Equality of these operator formulas does not yet identify C with C_* , or the Hodge circle of U_B with the formal action on U . It will let us constrain the unknown component of C using the small Ext^2 space. We now justify the actual formulas directly on the mirror, without a Hochschild comparison under HMS.

Use Todd-modified Hochschild–Kostant–Rosenberg isomorphisms. On homology the modification wedges with $\sqrt{td}$; on cohomology use the corresponding contraction modification. The module theorem of Calaque–Rossi–Van den Bergh identifies cap product with the polyvector action on Hodge cohomology [CRV12, Theorem 1.4]. It sends the Hochschild class of a perfect complex to its Mukai vector. We take normalized HKR, with the factor $1/p!$ in form degree p , and totalized form signs: holomorphic forms have Hochschild homological degree one; antiholomorphic cochains precede polyvector contractions, with the Koszul signs for moving each contraction through a cochain.

For this actual calculation, reuse the weight notation by choosing $t_+ \in H^{2,0}(B)$ and $t_- \in H^{0,2}(B)$ with $q(t_+, t_-) = 1$. Choose t_0 in the remaining Hodge line of U_B . These are complex Hodge directions, in place of the formal basis $1, \omega_Y, -[\mathrm{pt}]$ used above. The weight-zero space is now the full (p, p) -part of the Mukai space. The three summands

$$H^2(B, \mathcal{O}_B), \quad H^1(B, T_B), \quad H^0(B, \bigwedge^2 T_B)$$

act as (3.6), with x respectively in H^4 , $H^{1,1}$, and H^0 . Here T_B in the displayed groups is the tangent bundle. Wedge product gives the first action; contraction with a tangent-vector-valued one-form gives the middle derivation, skew for the Mukai form; bivector contraction gives the last action. These observations verify the relative signs in (3.6), up to an immaterial rescaling of each acting coordinate. The same form calculus on the abelian factor gives (3.7), now with a, b the Hodge directions of weights $1, -1$.

There are no cross terms involving the K3 factor, because its HH^1 vanishes. Consequently

$$\dim \mathrm{HH}^2(B \times E^g) = 22 + \binom{2g}{2} = b_2(Y \times T_g), \quad (3.8)$$

with the term 22 omitted in the torus-only case. Passing from Dolbeault-normalized to Betti-normalized classes does not alter these rank calculations. In an output component of type $(j, j+2)$, an acting polyvector of degree $b \in \{0, 1, 2\}$ uses the Mukai component of half-degree $j+b$. The powers of the comparison unit $\pm 2\pi i$ thus amount to invertible row and column rescalings.

Lemma 3.1. *Let $\mathcal{E}$ be supplied by Proposition 2.1, and let C be its normalized Mukai vector. The kernel of the operators (3.6)–(3.7) acting on C has dimension at least $\dim \mathcal{Z}$. If it has dimension exactly $\dim \mathcal{Z}$, the semiregularity trace on $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ is injective.*

Proof. The cap action on the class of an object factors as

$$\mathrm{HH}^2 \longrightarrow \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \mathrm{HH}_{-2}. \quad (3.9)$$

The first map is evaluation of a Hochschild operator on the object; the second is the boundary–bulk trace. Under HKR this trace is the collection of traces against powers of the Atiyah class, followed by an invertible Todd modification. These are the diagonal–adjunction and Atiyah–Chern character identities [Că105, BF08]. They apply to all the components of the trace, not only to $H^2(\mathcal{O})$. Rescaling the Mukai vector has no effect on rank.

By (3.8) and Proposition 2.1, the composite in (3.9) has rank at most $b_2(D) - \dim \mathcal{Z}$. This proves the kernel bound. If equality holds, the composite has rank equal to the upper bound on the dimension of its middle space. Both ranks must then equal that dimension, so the trace out of the middle space is injective. $\square$

3.3 Deforming a complex along its Hodge locus

We use the following precise form of semiregularity. It isolates the only deformation-theoretic input needed after the kernel calculations. The obstruction principle is due to Buchweitz–Flenner; the version for all infinitesimal obstructions and varying ambient varieties is also a consequence of Pridham’s theorem [BF03, Pri24].

Proposition 3.2. *Let $\mathcal{X} \rightarrow S$ be a smooth projective family over a smooth complex algebraic base, let $s \in S$, and let $\mathcal{E} \in \mathrm{Perf}(\mathcal{X}_s)$. Suppose the parallel transport of $\mathrm{ch}(\mathcal{E})$ remains Hodge on the germ of S at s , and that the map*

$$\mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \bigoplus_{j \geq 0} H^{j+2}(\mathcal{X}_s, \Omega^j), \quad o \longmapsto (\mathrm{Tr}(o \, \mathrm{At}(\mathcal{E})^j)/j!)_j$$

is injective. Then $\mathcal{E}$ extends to a compatible family of perfect complexes over the formal germ of S at s .

If S is connected and the Chern character is a global parallel section remaining Hodge on S , that section is algebraic on every fiber. The same conclusions hold for the Mukai vector in a family whose Todd classes are parallel algebraic classes.

Proof. We first explain the obstruction calculation. Resolve the initial perfect complex by a bounded complex of vector bundles, and use the Dolbeault resolution of its graded smooth bundle. A deformation is represented by a total-degree-one $(0, *)$ -superconnection squaring to zero. Relative smooth forms may be used over the Artin neighborhoods of s : in a smooth local trivialization take Taylor coefficients in base variables and set antiholomorphic base variables to zero. Local holomorphic coordinates identify these complexes with relative Dolbeault and de Rham resolutions. Smooth projectivity gives Hodge degeneration and cohomology base change.

Consider a small extension of Artin neighborhoods, with kernel I annihilated by the maximal ideal. Lift an already constructed superconnection arbitrarily, retaining the relative Dolbeault Leibniz rule. Its square o is an I -valued cocycle in the central endomorphism complex and represents the obstruction. Add a smooth $(1, 0)$ -connection and denote the curvature of the resulting full superconnection by R . The closed forms $\text{str}(R^m)$ represent the horizontal Chern character, up to the usual nonzero normalization; this follows from the superconnection Chern–Weil transgression formula.

Sort the curvature by holomorphic form degree. Its degree-zero part is o , and the other parts have degrees one and two. Since $I^2 = 0$, every term with two copies of o vanishes. Hence $\text{str}(R^m)$ has no component below holomorphic degree $m - 1$, and its component of degree $m - 1$ is

$$m \text{str}(o(R_0^1)^{m-1}), \quad (3.10)$$

where R_0^1 is the degree-one curvature at the central fiber. It represents the Atiyah class. The horizontal Chern character lies in filtration F^m by hypothesis. Hodge degeneration therefore makes (3.10) zero in Dolbeault cohomology. This remains true with coefficients in I on the central fiber, by freeness and cohomology base change. All semiregularity traces of the obstruction vanish. Injectivity makes o exact, so a correction of the lifted superconnection makes its square zero. Iterating over the powers of the maximal ideal gives compatible lifts.

These lifts are perfect complexes, not merely formal differential operators. Locally on a coordinate disk, assume inductively that the operator modulo I has been gauged to ordinary Dolbeault plus a holomorphic differential. Lift that gauge and an arbitrary holomorphic matrix lift of the differential. The difference has coefficients in I . Its highest positive form-degree component is Dolbeault closed by the square-zero equation. A local primitive gives a gauge transformation removing it. Continue downward in form degree. The remaining square-zero operator is ordinary Dolbeault plus a holomorphic differential. Thus it is locally the Dolbeault resolution of a perfect complex. The relative forms are flat over the Artin base, so the chosen reductions compute derived restriction. GAGA identifies these analytic lifts with algebraic perfect complexes.

The fixed bounded graded bundle in this local straightening gives a uniform finite Tor amplitude for the compatible lifts. Grothendieck existence algebraizes the resulting pseudocoherent tower over the completed local base [The26, Theorem 38.29.6, Tag 0DIG]. The family is proper, flat, and finitely presented, and the restriction to the closed fiber is perfect. The relative-perfectness criterion [The26, Lemma 37.71.3, Tag 0DJZ] therefore makes the algebraization perfect relative to the completed base. That base is regular and the family is smooth, so the total space is regular; the resulting bounded coherent complex is thus perfect. Spread its finite presentation to a finitely generated subalgebra of the completion over an affine neighborhood of s , retaining the closed point and its original complex. The resulting finite-type parameter is dominant over that neighborhood: the base is smooth at s , and its local ring maps injectively to its completion. On the relevant irreducible component, topological constancy of Chern characters identifies the classes of the spreading complex with the pullback of the prescribed parallel section. Its image contains a dense open subset of S .

For completeness, the last passage to every fiber uses proper cycle parameters, not constancy of a chosen complex. Express algebraicity of the fixed rational parallel class by finite sums and differences of cycles with a common positive denominator. Relative projective Hilbert schemes give countably many proper parameter spaces for such expressions; restrict to connected components on which the cycle class is the prescribed one. Their images in S are closed. A dense open subset of an irreducible complex variety cannot be covered by countably many proper closed subvarieties. One such proper image is therefore all of S . This proves algebraicity on every fiber. A connected smooth base is irreducible componentwise with no intersecting components, so the stated connected case

follows. Finally, multiplication by $\sqrt{\text{td}}$ is an invertible parallel algebraic operation in the indicated families and changes none of the conclusions. $\square$

The remaining task is now finite-dimensional: choose C_* so that the annihilator has the required wedge-injectivity, and show that the unknown Mukai component is forced by the cap-kernel bound. Once the bound is sharp, Proposition 3.2 transports the resulting algebraic tensor through the desired period family.

4 Exact Kuga–Satake correspondences

We now use the realization and deformation results to construct the prescribed Kuga–Satake map. The distinction between that map and an unspecified nonzero embedding matters: the full Clifford action will subsequently control tensor classes and the actions on different K3 factors.

For a projective K3 surface S , put

$$T(S) = \text{NS}(S)_{\mathbb{Q}}^{\perp} \subset H^2(S, \mathbb{Q}), \quad W_S = C^+(T(S), q),$$

where q is the cup form and our Clifford convention is $v^2 = q(v, v)$. Give W_S its standard left Kuga–Satake Hodge structure. Thus, for an oriented orthonormal basis u_1, u_2 of the positive period plane, its complex structure is left multiplication by $u_1 u_2$. This polarizable weight-one structure is $H^1(A_S, \mathbb{Q})$ for an abelian variety, up to the choice of lattice and isogeny [KS67, Del72, vG00]. If $w \in T(S)$ has nonzero square, the standard tensor is

$$j_w : T(S)(1) \longrightarrow \text{End}(W_S), \quad j_w(v)(x) = vxw. \quad (4.1)$$

A rational polarization identifies $W_S^{\vee}$ with $W_S(1)$ and therefore converts (4.1) into a Hodge map $\iota_w : T(S) \rightarrow W_S \otimes W_S \subset H^2(A_S^2, \mathbb{Q})$.

Theorem 4.1. *Let S be a smooth projective complex K3 surface. For every nonisotropic $w \in T(S)$, every full even-Clifford realization $H^1(A_S, \mathbb{Q}) = C^+(T(S), q)$, and every rational polarization used to define ι_w , there is a cycle*

$$\Gamma \in \text{CH}^2(S \times A_S^2)_{\mathbb{Q}} \quad \text{such that} \quad \Gamma_*|_{T(S)} = \iota_w.$$

The proof first constructs this exact map at the mirror quartic B . Auxiliary terms in its Mukai vector make the cap-action kernel rigid enough to determine the unknown transcendental component. Equality in the Ext bound then permits deformation through a full polarized K3 family. A rational comparison of polarization lines and specialization reach every projective K3 surface.

4.1 A spin-invariant tensor at the mirror quartic

Retain the spaces P_B and U from the cohomological dictionary. Choose a rational ample class $h \in \text{NS}(B)_{\mathbb{Q}}$ and set

$$N = h^{\perp} \cap \text{NS}(B)_{\mathbb{Q}}, \quad V = N \perp U.$$

Then $\dim N = 18$, while V has dimension 21 and signature $(2, 19)$. The square class of $q(h, h)$ can be prescribed as $2d$ for any positive integer d ; we explain this choice when comparing polarization degrees in Section 4.4.

For this V , write

$$W = C^+(V).$$

The left spin action on W admits a nondegenerate invariant alternating form that polarizes the Kuga–Satake structure on the chosen period component. One may use

$$E(x, y) = \text{Tr}(\alpha x^* y), \quad (4.2)$$

where $*$ reverses Clifford products and $\alpha^* = -\alpha$. For example, take α , up to sign, to be the product of two orthogonal rational vectors spanning a positive plane. At that plane, let I be their normalized product. Evaluating $\text{Tr}(Ix^*Ix)$ on the even Clifford monomial basis gives a definite form: commuting the two positive generators past a monomial compensates the parity of its negative generators. Choose the sign giving positivity. Spin invariance then gives positivity on the entire component.

Let $\theta \in \bigwedge^2 W$ be the nondegenerate alternating tensor corresponding to this form, and choose a rational volume element z in the full Clifford algebra of V . Define

$$J : V \longrightarrow \bigwedge^2 W, \quad J(v) = (L_{vz} \otimes 1)\theta. \quad (4.3)$$

Here L_a denotes left multiplication by a . Since $\dim V = 21$, the volume element is central and vz is reversal-even. Consequently L_{vz} is self-adjoint for (4.2), so (4.3) is alternating. It is also Spin-equivariant.

Restriction to the split group $\text{Spin}(U) = \text{SL}_2$ gives

$$W = M_0 \otimes H, \quad H = \mathbb{Q}e_+ \oplus \mathbb{Q}e_-.$$

Thus this is precisely a torus space in the preceding dictionary, with $g = \dim M_0 = 2^{19}$. To describe the operators on this large multiplicity space, work over $\mathbb{C}$. The even Clifford algebra in dimension 21 is the endomorphism algebra of its spin module Σ , of dimension 2^{10} . Its regular representation is therefore a sum of 2^{10} copies of Σ . On restriction to $N \perp U$, the module Σ is $M_1 \otimes H$, where M_1 is the 2^9 -dimensional irreducible module for the full Clifford algebra of $N_{\mathbb{C}}$. Consequently

$$W_{\mathbb{C}} = \Sigma \otimes R_0 = (M_1 \otimes H) \otimes R_0, \quad M_{0,\mathbb{C}} = M_1 \otimes R_0, \quad (4.4)$$

where R_0 is the regular-representation multiplicity space. The module Σ has an invariant alternating form. The tensor θ is the product of a symmetric nondegenerate tensor Q on M_0 with the alternating tensor on H . Moreover, Q factors as the form on M_1 times a symmetric form on R_0 .

For $y \in N$, the operator L_{yz} is, up to a common nonzero scale, the Clifford generator Γ_y on M_1 tensored with the identity on $R_0 \otimes H$. These generators are self-adjoint for Q . Let Δ be their chirality operator; it anticommutes with every Γ_y and is skew-adjoint, since reversal in dimension 18 has negative sign. For $u \in U$, L_{uz} is, up to a common scale, Δ times the vector action on H . It follows that $J(y)$ has symmetric matrix proportional to $\Gamma_y Q$ and alternating H factor, whereas $J(u)$ has matrix proportional to ΔQ and belongs to the $\text{Sym}^2 H$ part.

The Clifford generators on M_1 generate its full endomorphism algebra. An operator commuting with all of them can still act arbitrarily on R_0 . We remove this multiplicity freedom by choosing two further rational Spin-invariant nondegenerate alternating tensors θ_1, θ_2 whose relative operators to θ have scalar common commutant on R_0 . Such choices exist: over $\mathbb{C}$ two general symmetric operators relative to the multiplicity form have this property, which is a nonempty Zariski-open condition in the rational space of invariant tensors.

We next place these tensors in separate exterior degrees. This will let an annihilation equation distinguish the wedge, derivation, and contraction parts of a torus operator. Choose a rational basis

e_1, e_2, e_3 of $N^\perp \cap P_B$, each vector lying in one of $\mathbb{Q}h$, $H^0(B, \mathbb{Q})$, and $H^4(B, \mathbb{Q})$. For dual bases (v_j) and $(v_j^\vee)$ of V , put

$$\begin{aligned} R_1 &= 1, & R_2 &= \theta^4 + \theta_1^7 + \theta_2^{10}, & R_3 &= \theta^{13}, \\ C_* &= \sum_j v_j \otimes J(v_j^\vee) + \sum_{s=1}^3 e_s \otimes R_s. \end{aligned} \tag{4.5}$$

The first summand contains the desired Clifford map J . The three e_s components impose the additional equations just described; after those equations remove the unwanted operators, the remaining kernel will consist of the 19 simultaneous spin lowering directions. We prove this next. The full tensor is invariant under the simultaneous SL_2 action. Its inverse image under the dictionary is therefore a rational middle-degree class a killed by the symplectic form.

4.2 The formal kernel has dimension nineteen

For a tensor in $U \otimes \bigwedge W$, write its three weight components as

$$t_+ \otimes U_- + t_0 \otimes U_0 + t_- \otimes U_+. \tag{4.6}$$

Here $U_-, U_0, U_+ \in \bigwedge W$. A superscript $*$ denotes the components of C_* ; we will later use the same notation for the component in $U_B \otimes \bigwedge W$. In particular, U_-^* is nonzero and has exterior degree two. The annihilator of a by degree-two classes is computed by the operators $D_x + F(\lambda, L, \pi)$ in (3.6) and (3.7).

Lemma 4.2. *For the class a defined by (4.5), its annihilator $\mathcal{Z} \subset H^2(Y \times T_g, \mathbb{Q})$ has dimension 19 and projects injectively to $H^2(Y, \mathbb{Q})$. In particular, it satisfies (2.4).*

Proof. Decompose $x = x_N + x_e + x_0 t_0$, where $x_e = \sum_s x_{e,s} e_s$. The e_s components of the annihilation equation are

$$F(R_s) + x_{e,s} U_-^* = 0. \tag{4.7}$$

For $s = 2$, the second term has exterior degree two. On a power θ_i^k , the contraction, derivation, and wedge terms of F have degrees $2k - 2, 2k, 2k + 2$, respectively. For $k = 4, 7, 10$ these nine degrees are distinct and exceed two. Hence $x_{e,2} = 0$, and each of the nine terms vanishes separately.

Symplectic Lefschetz injectivity now gives $\lambda = 0$ from $\lambda \wedge \theta^4 = 0$. The derivation terms give $F_L(\theta_i) = 0$ for $i = 0, 1, 2$, where $\theta_0 = \theta$. For the contraction term, contracting two a directions into θ^4 gives, up to a nonzero factor, the two- b tensor obtained from π by Q , wedged with θ^2 . The same injectivity gives $\pi = 0$. All these wedge maps are in the injective range for $g = 2^{19}$. The remaining equations (4.7) give $x_e = 0$.

The equation $F_L(\theta) = 0$ says that L is self-adjoint for Q . Preservation of θ_1 and θ_2 says in addition that L commutes with their relative operators. If $x = 0$, the N equations become $F_L(J(y)) = 0$ for every $y \in N$. By the matrix description above, these are exactly $[L, \Gamma_y] = 0$. Irreducibility on M_1 , followed by the common-commutant condition on R_0 , makes L scalar. The equation $F_L(U_0^*) = 0$ kills this scalar, because the lowering action on the highest-weight-two representation is nonzero. The kernel therefore projects injectively to $N \oplus \mathbb{Q}t_0$, a space of dimension 19.

Conversely, each $x \in N \oplus \mathbb{Q}t_0$ occurs in the kernel. The operator D_x is the lowering-degree element of $\mathfrak{so}(V)$ displayed in (3.6). Its simultaneous spin action preserves (4.5). Since W has only Cartan weights 1 and -1 , its lowering action is of the form F_L . This proves the dimension assertion. Finally, projection of $H^3(Y \times T_g, \mathbb{Q})$ to $H^2(Y, \mathbb{Q}) \otimes H^1(T_g, \mathbb{Q})$ identifies the map in (2.4) with the tensor product of the injective projection of $\mathcal{Z}$ and the identity. $\square$

4.3 Determining the transcendental component

Proposition 2.1 now supplies a perfect complex on $B \times E^g$. Divide its Mukai vector by the common nonzero rational scalar in the Euler comparison and call the result C . The dictionary determines $C|_{P_B} = C_*|_{P_B}$, but does not initially determine its component on $U_B = T(B)$. The Ext bound does. In the following statement, give the formal space U the Hodge structure induced by the actual elliptic structure on H through its highest-weight-two representation.

Lemma 4.3. *There is a rational Hodge isometry $U_B \simeq U$, for the actual Hodge structure on $H = H^1(E, \mathbb{Q})$, under which $C = C_*$. The semiregularity trace of the perfect complex is injective.*

Proof. Excluding a zero transcendental component. Choose normalized actual Hodge directions t_+, t_0, t_- on U_B , as in the cap calculation of Section 3, and write its unknown component in the form (4.6). By Lemma 4.2 and the Ext bound, the actual cap kernel has dimension at least 19. Imposing the three linear conditions $x_e = 0$ leaves a subspace of dimension at least 16. On this subspace the equations $F(R_s) = 0$ again imply $\lambda = \pi = 0$, and make L self-adjoint and commuting with the two relative operators. The N equations are

$$F_L(J(y)) + q(x_N, y)U_- = 0 \quad (y \in N). \quad (4.8)$$

The t_- component, using (3.6), is

$$F_L(U_+) = J(x_N) + q(x_0 t_0, t_0)U_0. \quad (4.9)$$

If the entire component (4.6) were zero, Equation (4.8) would make L scalar, and (4.9) would give $J(x_N) = 0$, hence $x_N = 0$. Only the scalar L and x_0 would remain, contrary to the dimension lower bound 16. Thus the component is nonzero.

Identifying the Hodge representation. This nonzero component realizes the simple, three-dimensional K3 Hodge structure U_B in the exterior algebra of copies of H , with the indicated Tate twists. Since E is non-CM, its irreducible tensor representations are the irreducible SL_2 representations. The only possible one here has highest weight two. The resulting rational Hodge identification $U_B \simeq U$ is a similarity: the invariant forms on this irreducible representation are proportional. The underlying quadratic spaces are already rationally isometric by the dictionary. In odd dimension their similarity multiplier is a rational square, as follows by taking determinants; the signatures give the positive sign. Rescaling gives a Hodge isometry. We may now use it to compare the two sets of weight directions. The component (4.6) is equivariant for the simultaneous SL_2 ; in particular $U_- \neq 0$.

Determining the component up to one scalar. Still on the subspace $x_e = 0$, the kernel of projection to x_N has dimension at most two: setting $x_N = 0$ in (4.8) again makes L scalar, leaving at most that scalar and x_0 . The image in $N_{\mathbb{C}}$ therefore has dimension at least 14. A totally isotropic subspace of the 18-dimensional nondegenerate space $N_{\mathbb{C}}$ has dimension at most nine. Choose a kernel element for which $q(x_N, x_N) \neq 0$.

For every $y \perp x_N$, Equation (4.8) says that L commutes with Γ_y . On the irreducible Clifford module M_1 , the commutant of these 17 directions is spanned by 1 and $\Delta\Gamma_{x_N}$; this is the restriction of the even-dimensional Clifford module to an odd-dimensional perpendicular. The relative operators again remove the multiplicity commutant, so

$$L = l1 + m\Delta\Gamma_{x_N}.$$

The Clifford relation gives

$$[\Delta\Gamma_{x_N}, \Gamma_y] = \kappa q(x_N, y)\Delta$$

for a fixed nonzero normalization constant κ . Equation (4.8) therefore makes U_- a nonzero multiple cU_-^* : the two- b tensor with matrix ΔQ is exactly the lowest-weight line of $J(U)$. Equivariance gives this same scalar on all three components in (4.6).

Determining the exact normalization. Let L^* be the spin lowering lift for $x = x_N$ in the tensor C_* . Comparing its version of (4.8) with the actual equation gives

$$L = cL^* + \tilde{l}1.$$

Now use Equation (4.9) modulo the line of U_0 . The scalar part $\tilde{l}1$ of L lowers U_+ into that line. The equation for C_* , with $x = x_N$, gives $F_{L^*}(U_+^*) = J(x_N)$. Since $U_+ = cU_+^*$, we obtain

$$J(x_N) \equiv F_L(U_+) \equiv c^2 F_{L^*}(U_+^*) = c^2 J(x_N) \pmod{\mathbb{C}U_0}.$$

One factor c comes from L and the other from U_+ . The tensor $J(x_N)$ lies outside that line, since $J(N)$ and $J(U)$ have the distinct Clifford descriptions above. Hence $c^2 = 1$. Reversing the sign of the transcendental Hodge isometry if necessary gives $C = C_*$.

The circle on the resulting $V = h^\perp$ has its unique lift to the spin group, and its action on W is the actual circle on $H^1(E^g, \mathbb{Q})$. Thus this actual weight-one structure is the stipulated Kuga–Satake structure. The formal and actual cap calculations now agree, so the cap kernel has dimension exactly 19. Lemma 3.1 makes the semiregularity trace injective. $\square$

4.4 Deformation and all polarization degrees

We have constructed an exact tensor whose complex can deform whenever its Mukai vector stays Hodge. We first use this to move away from B while preserving its polarization degree. We then compare that degree with any prescribed degree by a rational isometry, and specialize the exact tensor to fibers with larger Picard rank.

Deformation from the mirror quartic. Consider the full family of h -polarized K3 surfaces, coupled with their Kuga–Satake abelian varieties. These are algebraic families after a suitable level cover. More precisely, the primitive period domain and its spin cover, together with the invariant rational alternating form, give a holomorphic equivariant map to Siegel space. Arithmetic subgroups under the isogeny from Spin to SO have arithmetic image. Intersect with the stabilizers of the chosen lattices and pass to a neat level; the K3 period theorem and algebraicity of the arithmetic quotients and their morphisms then give a smooth algebraic base with both universal families [Del72, BB66, Bor72].

The lattice at the central point may be chosen to give exactly the abelian variety E^g : preserve that lattice by further level restriction and multiply the rational polarization by an integer, with the sign chosen above. On this spin level cover the tensors in (4.5) are global parallel sections: the vectors h , H^0 , and H^4 remain fixed and algebraic, the sum on V is Spin-invariant, and every θ_i is Spin-invariant. Hence the full Mukai vector C_* stays Hodge throughout the family; so do its Todd factors. Lemma 4.3 and Proposition 3.2 give algebraicity of this class on every fiber of the polarized component.

On a fiber $S \times A$, take the component in $H^2(S) \otimes H^2(A)$ and project the K3 factor to $V = h^\perp$. The additional terms $e_s \otimes R_s$ have abelian degrees 0, 8, 14, 20, or 26, so these algebraic projections extract J . To regard J as a correspondence with two H^1 slots, use the surface and abelian Künneth projectors, the polarization and its inverse, and the inclusion of an exterior square into two ordered slots. All are algebraic: the inverse polarization is provided by the Lefschetz standard conjecture for abelian varieties, and the slot conversion can also be realized using the addition map and pullback.

This already determines the usual standard tensor, not only a tensor with the volume element on the right. Since z is central, $L_{vz} = L_v R_z$ on the even space. For any nonisotropic $w \in V$, replace R_z by R_w using right multiplication by the invertible even element $z^{-1}w$. Every right even multiplication is a weight-one Hodge endomorphism of the abelian variety, and is therefore algebraic. Thus the map $x \mapsto vxw$ is algebraic in this full primitive Clifford construction.

Comparison of polarization degrees. Fix a prescribed degree $2d$. The initial rational ample class h on B can be chosen with square $2d$. Indeed, Meyer’s theorem gives an isotropic rational vector in the indefinite rank-19 space $\text{NS}(B)_{\mathbb{Q}}$, hence a rational hyperbolic plane. The quadric $q(x, x) = 2d$ consequently has rational points, and rational approximation on this quadric gives a point in the open ample cone. Scaling to a primitive integral class changes its square by a rational square. Thus the primitive integral class chosen on B has square $2d$ times a rational square. Witt’s extension theorem therefore gives a rational isometry of the K3 quadratic spaces carrying its polarization line to the line of a primitive vector of square $2d$. The isometry identifies the primitive period domains. At very general periods on both sides the surfaces have Picard rank one: the excluded wall hyperplanes, including their translates under the rational isometry, are rational hyperplanes and can all be avoided. The K3 period theorem supplies both polarized surfaces. Buskin’s theorem makes their rational Hodge isometry algebraic [Bus19]. If f denotes its restriction to the primitive spaces, the induced Clifford Hodge isomorphism satisfies

$$C^+(f) j_w(v) C^+(f)^{-1} = j_{f(w)}(f(v)).$$

This weight-one isomorphism is induced by an abelian isogeny up to a rational multiple. Composing it with the K3 correspondence, and converting polarizations as above, therefore transports the exact tensor. We obtain the exact standard tensor very generally in degree $2d$. One may make this comparison on marked components or on commensurable level quotients of the two domains.

Specialization and transcendental restriction. Specialization gives the full primitive tensor at every polarized fiber. Here the class being specialized is the prescribed parallel tensor in the full Kuga–Satake family, so Picard jumps cause no loss of information. Equivalently, apply the specialization theorem and its primitive-map corollary in [Ope26a, Theorem 1.2 and Corollary 5.5]: the very-general nonzero algebraic input required there has just been supplied in its stronger exact form. This specialization is the proper-cycle-parameter argument of Proposition 3.2; it does not replace actual algebraicity by the Hodge condition.

Proof of Theorem 4.1. Every projective K3 surface has a primitive ample class, so the preceding construction gives its full primitive Kuga–Satake tensor. The projection to $T(S)$ is algebraic, using the divisor intersection pairing and the degree-zero and top-degree projectors. For the prescribed $w \in T(S)$, restrict the even Clifford space to

$$C^+(T(S)) \subset C^+(h^{\perp}).$$

This is a rational weight-one sub-Hodge structure, and $x \mapsto vxw$ preserves it for $v \in T(S)$. It can therefore be split off by algebraic abelian correspondences. Changing its lattice, isogenous realization, or rational polarization also uses algebraic abelian correspondences. The resulting cycle has the exact action (4.1) with the chosen polarization conversion, proving the theorem. $\square$

Corollary 4.4. *For every projective complex K3 surface S , the rational Hodge conjecture holds in every degree on every self-power S^m .*

Proof. Theorem 4.1 supplies the exact standard tensor on the full even Clifford space, for one nonisotropic w and one polarization. These are the hypotheses of the general self-power criterion [Ope26b, Theorem 1.1]. That criterion is independent of the special quadratic locus used elsewhere in the same reference to construct correspondences. $\square$

5 Degree-four tensors in spin families

The exact Kuga–Satake correspondences control each K3 surface separately. To compare several factors, we will also need a quadratic contraction tensor on certain abelian varieties. The construction below permits independent period planes at every real place of a totally real field. Its four weight-one slots will later compare actions on different spin constituents.

Let $F'/\mathbb{Q}$ be a totally real Galois extension and let (V', q') be a quadratic space over F' of dimension d' . Assume that

$$d' \geq 5, \quad d' \equiv 5 \pmod{8}, \quad \text{sig}(V'_\sigma) = (2, d' - 2) \quad (\sigma : F' \hookrightarrow \mathbb{R}), \quad \text{Witt}(V') \geq 2. \quad (5.1)$$

The dimension can be increased as needed in the application. Choose an oriented positive period plane at each real place. The corresponding spin circles give the weight-one Hodge structure

$$W' = \text{Res}_{F'/\mathbb{Q}} C_{F'}^+(V'). \quad (5.2)$$

It is polarizable: choose an invariant alternating form with the required positivity at every real embedding, using approximation over F' in the construction (4.2), and take its trace to $\mathbb{Q}$. Thus W' is the rational first cohomology of an abelian variety A' , with a chosen lattice. On each component the period construction gives an algebraic family of abelian varieties. Indeed, a neat arithmetic subgroup of the restriction-of-scalars spin group can be chosen to preserve the lattice and polarization. The product of the orthogonal period domains then maps holomorphically and equivariantly to Siegel space, and the algebraicity theorems used in Section 4.4 apply.

Choose a volume element z' over F' . Define a rational tensor

$$\mathcal{P} \in (\text{End } W')^{\otimes 2} \quad (5.3)$$

by prescribing its extension to $\mathbb{C}$. For each embedding $\sigma : F' \hookrightarrow \mathbb{C}$, take dual bases (v_j) and $(v_j^\vee)$ of V'_σ and form

$$\sum_j L_{v_j z'} \otimes L_{v_j^\vee z'}.$$

Each operator is supported on the σ summand of $W'_\mathbb{C}$; sum these tensors over σ . This is the trace construction, so it descends to $\mathbb{Q}$. It is invariant under $\text{Res}_{F'/\mathbb{Q}} \text{Spin}(V')$. Using an invariant polarization in each endomorphism factor, regard $\mathcal{P}$ as a degree-four Hodge tensor in four ordered $H^1(A')$ slots.

Proposition 5.1. *For the data (5.1), the tensor $\mathcal{P}$ is the class of a cycle in $\text{CH}^2((A')^4)_\mathbb{Q}$, for every choice of the period planes and every isogenous realization of (5.2).*

Proof. We first choose a point of the family whose abelian variety is a power of the non-CM elliptic curve E in the torus dictionary. The Witt-index assumption gives a split rank-three quadratic subspace of V' with signature $(2, 1)$ at every real place: take one hyperbolic plane and a vector positive at all places in a second one. Its spin group is SL_2 over F' . Restricting scalars and then taking the diagonal SL_2 over $\mathbb{Q}$ makes W' a multiple of its standard representation.

The common circle can be placed in any prescribed component of the period family. Under the ternary identification, conjugate by an element of $\mathrm{GL}_2(F')$ with the chosen determinant signs at the real embeddings. Approximation permits these signs to be chosen independently. This also accommodates either elliptic marking convention in the graph dictionary; its common line sign was fixed before the choice of tensor. Choose the lattice to give the corresponding elliptic power exactly at this point.

Apply the torus-only realization with

$$W = (W')^{\oplus 4} = M_0 \otimes H,$$

using bases compatible with the four copies. Place $\mathcal{P}$ in $\bigwedge^4 W$, with one vector in each copy, after the polarization conversion. These copies have a second use: to verify the wedge-injectivity hypothesis, we will project a degree-one class from one copy against an annihilator's degree-two component on another. We arrange that the annihilator has the same component on every copy.

Let θ be the direct sum of four identical copies of a nondegenerate rational Spin-invariant alternating tensor on W' . Choose two more tensors θ_1, θ_2 by tensoring that tensor with nondegenerate symmetric matrices on the four copy indices: the first is diagonal with distinct entries, and the second has every entry nonzero. Here the relative operators act on the four copy indices. Their commutation conditions will force identical blocks there, while allowing operators within W' ; this is the copy condition just required. As in the earlier construction, separate the powers so that wedge, derivation, and contraction terms have different degrees. Put

$$C_* = \mathcal{P} + \theta^6 + \theta_1^9 + \theta_2^{12}. \quad (5.4)$$

This is Spin-invariant and hence invariant under the chosen simultaneous SL_2 . The torus dictionary produces a middle-degree primitive class a . Here

$$g = \frac{1}{2} \dim_{\mathbb{Q}} W = [F' : \mathbb{Q}] 2^{d'} \geq 32,$$

so g is even, the realization dimension condition is satisfied, and the symplectic wedge maps used below are in their injective range.

We verify the wedge-injectivity hypothesis of Proposition 2.1. In the formal annihilation equation $F(\lambda, L, \pi)C_* = 0$, the three operator types on the displayed powers have degrees $(10, 12, 14)$, $(16, 18, 20)$, and $(22, 24, 26)$. They are all distinct and exceed the degrees 2, 4, 6 obtained by acting on $\mathcal{P}$. The same symplectic wedge and contraction calculation as in Lemma 4.2 therefore gives $\lambda = \pi = 0$ and

$$F_L(\theta) = F_L(\theta_1) = F_L(\theta_2) = 0.$$

The first condition makes L self-adjoint. Commutation with the diagonal relative operator makes it block diagonal on the four copies. Commutation with the second relative operator, whose off-diagonal entries are nonzero, then makes all four blocks equal.

Under the inverse dictionary, the annihilator $\mathcal{Z}$ consequently consists of degree-two forms obtained by repeating the same form on each of the four torus copies. Its projection to the degree-two cohomology of any one copy is injective. For an input in H^1 of copy i , choose $j \neq i$ and project its wedge product to multidegree one on copy i and degree two on copy j . This is the tensor product of the identity on H^1 of copy i with that injective projection of $\mathcal{Z}$. Inputs from the other copies have different multidegrees. Taking these projections for all i proves (2.4).

In the torus-only case, the Euler tests determine the entire normalized Mukai vector, so the resulting perfect complex has $C = C_*$. At this elliptic point, the actual cap kernel and the formal annihilator have the same dimension: the two sets of weights on this invariant tensor are conjugate

over $\mathbb{C}$. The Ext bound is therefore an equality at the cap rank, and the semiregularity trace is injective.

Throughout the indicated algebraic period family, all terms of (5.4) remain parallel Hodge classes by Spin invariance. Apply Proposition 3.2 and extract cohomological degree four. The three higher powers have degrees 12, 18, and 24, so the extracted class is exactly $\mathcal{P}$. Its one-slot-per-copy component gives an algebraic class on $(A')^4$. The elliptic starting point was available in every component. Finally, transport under abelian isogenies and change of rational polarization use algebraic correspondences, proving the assertion for all the stipulated realizations. $\square$

6 Hodge classes on mixed products

Fix projective K3 surfaces $S_1, \dots, S_m$, and put

$$V_i = T(S_i)(1), \quad W_i = C^+(T(S_i)).$$

Here V_i has weight zero, whereas W_i carries the weight-one Kuga–Satake structure. Theorem 4.1 supplies the exact Clifford correspondences, and Corollary 4.4 supplies all Hodge classes on each individual self-power. We must still treat tensors involving several different V_i .

The individual assertions determine the projections of the joint Hodge group, but not the relations between those projections. We first use weight-one Hodge morphisms between the Kuga–Satake varieties to compare the actions of shared simple factors projectively. The CM theorem then controls the torus action. After removing these actions, only certain central signs remain, arising from the two half-spin representations in dimension four. We encode their compatibility by a finite graph and use the degree-four tensors of Proposition 5.1 to prove the required relations. All auxiliary abelian varieties will be chosen before choosing the tensor automorphism to which this comparison is applied.

6.1 Tensor groups and the individual factors

For a finite collection of the surfaces and abelian varieties used here, let G be its Hodge group, with Tate objects fixed. Let C be the group of tensor automorphisms of its cohomology realizations that preserve algebraic correspondences, again fixing Tate objects. Then $G \subset C$. We use these groups over $\mathbb{Q}$ and extend their coefficients to $\mathbb{C}$ when decomposing representations.

We recall why C has the required invariant interpretation. The Lefschetz standard conjecture holds for surfaces and abelian varieties and is stable under products; it gives algebraic Künneth and Lefschetz operators in the category under consideration [Kle94]. Hodge–Riemann then identifies numerical and homological equivalence there. Numerical semisimplicity [Jan92], with the symmetry modified to give the ordinary flip on cohomology, gives a neutral Tannakian category of pure motives with its cohomology fiber functor [DM82]. Its tensor automorphism group is the group just described.

In weight zero, the C -invariants are precisely algebraic tensors. Indeed, before fixing Tate objects, invariants are morphisms from the unit motive. The central weight torus acts by cohomological weight and maps onto the group acting on Tate objects up to isogeny. It acts trivially in weight zero, so fixing Tate objects does not change those invariants. Consequently it will suffice to prove

$$C|_{\bigoplus_i V_i} = G|_{\bigoplus_i V_i} \tag{6.1}$$

for one finite collection containing the original surfaces and the auxiliary abelian varieties constructed below.

Lemma 6.1 (Individual actions). *For any such collection and any $c \in C(\mathbb{C})$, the action of c on V_i belongs to $\text{Hg}(V_i)(\mathbb{C})$. Its simultaneous action on V_i, W_i is induced by a lift to the spin group of the quadratic space V_i .*

Proof. Corollary 4.4, equivalently the exact-tensor criterion of [Ope26b, Theorem 1.1] applied after Theorem 4.1, makes every Hodge tensor on powers of V_i algebraic. Algebraic degree projectors and the quadratic pairing allow the usual tensor criterion for the Hodge group. Hence $c|_{V_i}$ lies in $\text{Hg}(V_i)$.

Choose a spin lift of this orthogonal action. The operator it induces on W_i preserves the exact KS tensor, all right even Clifford multiplications, and an invariant polarization. These tensors are also preserved by c : the KS tensor is algebraic, while the right multiplications are weight-one Hodge endomorphisms of an abelian variety and therefore algebraic. After removing the chosen spin lift, the remaining operator on W_i commutes with all right multiplications, so it is left multiplication by an even Clifford element a . Preservation of the KS map

$$v \mapsto L_v R_w, \quad w^2 \neq 0,$$

then says that a commutes with every $v \in V_i$. The even part of the center of the full Clifford algebra consists of scalars. Thus the remaining operator is scalar, and preservation of the polarization makes it 1 or -1 . Both scalars are supplied by the spin cover. $\square$

Let

$$G_0 = \text{Hg} \left(\bigoplus_i V_i \oplus \bigoplus_i W_i \right).$$

The projection from G_0 to the Hodge group of the V_i has finite kernel: the Hodge circle acts on each W_i through the spin lift of its action on V_i . Thus the W_i do not add a Lie algebra factor.

For each i , write $E_i = \text{End}_{\text{Hdg}}(V_i)$. Zarhin's theorem says that E_i is either totally real or CM [Zar83]. We use the following notation:

- in the orthogonal case, $F_i = E_i$ and $d_i = \dim_{E_i} V_i \geq 3$;
- in the unitary case, $F_i = E_i^+$ is the maximal totally real subfield and $d_i = \dim_{E_i} V_i \geq 1$.

Over $\mathbb{C}$, the corresponding groups and representations are

	$\text{Hg}(V_i)_{\mathbb{C}}$	$(V_i)_{\mathbb{C}}$	
orthogonal	$\prod_{p: F_i \hookrightarrow \mathbb{C}} \text{SO}_{d_i}$	one standard space at each p	(6.2)
unitary	$\prod_{p: F_i \hookrightarrow \mathbb{C}} \text{GL}_{d_i}$	a standard space and its dual at each p .	

The quadratic form in the orthogonal case, and the underlying quadratic form of the Hermitian form in the unitary case, has positive index two at the distinguished real embedding and is negative definite at the other real embeddings. We call the distinguished embedding the *active place*. The infinitesimal Hodge circle has weights 2, -2 once each on V_i , and weight zero elsewhere.

6.2 Shared simple ideals and spin constituents

The derived Lie algebra of G_0 is a subdirect product of the simple ideals occurring in (6.2). It is therefore a product of simple ideals, each mapping diagonally onto some of the factor ideals. We call two factor ideals *shared* if they are images of the same ideal of $\text{Lie}(G_0)^{\text{der}}$.

Lemma 6.2 (Possible identifications). *Shared simple ideals have one of the following forms:*

1. *orthogonal factors of the same dimension, with their standard modules matched;*
2. *unitary factors of the same dimension, with their standard modules matched up to duality;*
3. *ideals isomorphic to $\mathfrak{sl}_2$, shared among orthogonal factors of dimensions three or four and unitary factors of dimension two.*

Whenever sharing occurs, the fields F_i , viewed in $\mathbb{R}$ through their active embeddings, agree. The associated place identifications are then the identity on that common field. In the unitary case with $d_i \geq 3$, matching the standard modules as representations of the shared simple ideal also matches their characters on the connected center of G_0 .

Proof. An identification of shared ideals must identify their infinitesimal Hodge-circle elements. Their eigenvalues first restrict the possible representations; the Galois translates of the nonzero components will then identify their fields and places. First transport a shared ideal by Galois to an active place. In an orthogonal standard module the circle element rotates one plane. For $\mathfrak{so}_4 = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, both simple ideals are active, with weights $1, -1$ on their standard spin modules. The spin standard module for $\mathfrak{so}_3$ has the same weights. In an active unitary standard module there is exactly one nonzero weight, equal to 2 or -2 ; the simple ideal sees its traceless part.

These elements distinguish the possible identifications of the simple Lie algebras. In the accidental isomorphism $\mathfrak{so}_6 \simeq \mathfrak{sl}_4$, the orthogonal half-spin module has weights $1, 1, -1, -1$, whereas a unitary standard module has the traceless pattern of a single nonzero weight. They cannot match. In type D_4 , a triality automorphism carrying the standard module to a half-spin module is likewise excluded: the latter has only weights $1, -1$, unlike the orthogonal standard module. The remaining coincidences are precisely those in the statement.

For completeness, the same argument identifies the fields and the place labels. Work with simple ideals over $\overline{\mathbb{Q}}$, testing activity after the fixed embedding into $\mathbb{C}$. For an ideal at a place p of F_i , the Galois elements carrying it to an active ideal are exactly those carrying p to the active place of F_i . This holds also in dimension four, since both ideals there are active. The sets agree for shared ideals. After transporting one shared ideal to the active place, equality of these sets is equality of the stabilizers of the active field embeddings. Thus the embedded fields agree, and all place matchings are the matching embeddings of that field. Sharing is transported to every such place; for an orthogonal dimension-four block this may initially involve only one of its two simple ideals.

Finally consider a shared unitary ideal of dimension $d \geq 3$ and choose the standard or its dual so that the simple modules agree. The exceptional eigenvalue in the traceless circle element determines the sign of its scalar part. Hence the central characters of the two matched modules have the same differential on the central component of the circle. This is true after every Galois transport. A Galois stable character condition that vanishes on the circle vanishes on the rational torus generated by its projection. Applying this to the difference of the two characters proves equality on the connected center. One may pass to powers of characters to descend to the central torus quotient; its character lattice is torsion free, so equality of these powers implies equality of the original characters. For $d = 2$, the traceless pattern does not distinguish the sign, which is why that case remains in the third alternative. $\square$

We will use the constituents of W_i as representations of the joint group. Restriction of the full even Clifford representation to an orthogonal sum gives all choices of spin constituents. In the orthogonal case, these are tensor products over the places of F_i : there is one spin choice for an odd-dimensional block and either half-spin choice for an even-dimensional block. Multiplicities will

not matter. In dimension three the choice is the standard module of one SL_2 ; in dimension four the two choices are the standard modules of the two SL_2 factors.

In the unitary case the choices at a place are

$$\bigwedge^r \mathrm{Std} \otimes (\det)^{-1/2}, \quad 0 \leq r \leq d_i, \quad (6.3)$$

with independent choices at the different places. The half-determinant notation denotes a character on the relevant spin cover. When $d_i = 2$, these choices are the standard module of SL_2 , with zero connected central weight, and two opposite torus characters. These descriptions follow from the exterior-algebra model of spin and the spin decomposition of an orthogonal sum.

Lemma 6.3 (Weight-one comparison). *If a simple ideal is shared between factors i and j , then W_i and W_j contain isomorphic representations of the joint Hodge group on which that ideal acts nontrivially. The required intertwiners and the projections onto individual copies of these representations are algebraic correspondences with complex coefficients.*

Proof. For a shared $\mathfrak{sl}_2$, choose its standard module at the given place and a transported shared standard module at every other place. Lemma 6.2 provides the common field and these transported identifications. In the other orthogonal types, choose matching spin constituents at all places. In the unitary types, choose matching constituents in (6.3); their central characters agree by Lemma 6.2. Thus the tensor constituents have the same joint Lie algebra representation. The Hodge group is connected, so they have the same group representation.

Morphisms between weight-one Hodge structures of complex abelian varieties are induced by rational homomorphisms of abelian varieties. After extending coefficients, every intertwiner just used is therefore algebraic. The same applies to the weight-one idempotents selecting the constituents and individual multiplicity copies. $\square$

6.3 Removing projective and torus actions

We prove the reduction before constructing the auxiliary varieties. For this subsection, allow any finite collection containing the original surfaces and their W_i whose Hodge group G projects onto G_0 with finite kernel. Write C for its algebraic tensor group as above, and identify the Lie algebras of G and G_0 . The auxiliary varieties constructed below will satisfy this hypothesis. Thus the reduction will apply to the final collection once all of its objects have been fixed.

Lemma 6.4 (Projective compatibility). *For $c \in C(\mathbb{C})$, its actions at all occurrences of a shared simple ideal agree in the corresponding adjoint group.*

Proof. By Lemma 6.1, c acts on each W_i by a spin lift of its action on V_i . This has the tensor-product description of its block spin constituents, up to an overall sign. Lemma 6.3 provides matching constituents on which a specified shared ideal acts nontrivially. Their intertwiner is algebraic, so c has the same action in these two realizations. Equality of invertible tensor-product operators forces equality of their individual factors up to scalars. A nontrivial representation of a simple Lie algebra is projectively faithful on its adjoint group. Hence the adjoint actions at the shared ideal agree. $\square$

Let W_{CM} be the sum, in all the weight-one realizations of the collection under consideration, of the constituents on which $\mathrm{Lie}(G)^{\mathrm{der}}$ acts trivially. This is a rational sub-Hodge structure. It is the realization of a CM abelian factor, and its projector is algebraic. The CM Hodge conjecture [Ope26d, Theorem 1.1], applied jointly to these factors and their powers, therefore gives

$$C|_{W_{\mathrm{CM}}} = G|_{W_{\mathrm{CM}}}. \quad (6.4)$$

Lemma 6.5 (Reduction to central actions). *After multiplying $c \in C(\mathbb{C})$ by an element of $G(\mathbb{C})$, one may assume that its compatible adjoint actions through the original factors are trivial at every simple ideal and that it acts as the identity on W_{CM} . Under these assumptions all actions on the V_i outside the dimension-four orthogonal blocks and the dimension-two unitary blocks can be removed by centers of the shared simply connected derived factors.*

Proof. Lemma 6.4 gives a single prescribed action in the product of the adjoint groups, and (6.4) gives the prescribed action on the CM part. The derived group covers the first product, and the connected center covers the torus image on W_{CM} . These choices can be made independently. Removing their product proves the first assertion. Continue to write c for the resulting element.

On an odd-dimensional orthogonal block with trivial adjoint action, the standard action is already the identity. On an even-dimensional orthogonal block it is a sign. In a spin lift this sign is the ratio of the scalar actions on the two half-spin choices. For simple types other than $\mathfrak{sl}_2$, these ratios agree at shared ideals: apply the weight-one intertwiners to matching tensor words with either half-spin choice at the specified place. Interchanging the two choices in one description only inverts the ratio and does not change a sign. The center of the shared simply connected factor supplies that standard-module sign, so it can be removed.

On a unitary block at (i, p) , choose the standard or its dual so that the simple modules match on each shared ideal, as in Lemma 6.2. Write its scalar action as $r_{i,p}$; the opposite choice has action $r_{i,p}^{-1}$. In (6.3), compare choices 0 and d_i at that place, using only these two choices at every other place. Both resulting words belong to W_{CM} , and their ratio is $r_{i,p}^{d_i}$. Hence

$$r_{i,p}^{d_i} = 1. \quad (6.5)$$

This removes the case $d_i = 1$. For $d_i \geq 3$, compare also the choice 1 with the choice 0. The ratio is $r_{i,p}$, and the matched weight-one constituents show that these ratios agree on shared simple ideals. They are therefore supplied by the center of the shared SL_{d_i} and can be removed there. Such derived central corrections do not affect W_{CM} . The only remaining actions on the V_i are therefore signs on orthogonal dimension-four and unitary dimension-two blocks. $\square$

6.4 The graph and its auxiliary abelian varieties

We now formulate the compatibility needed for these residual signs and construct algebraic tensors that enforce it. The graph and all its auxiliary varieties depend only on the original Hodge representations, before any element of the final group C is chosen.

Form a finite graph Γ as follows. Its ordinary vertices are the simple ideals of $\text{Lie}(G_0)_{\mathbb{C}}^{\text{der}}$ isomorphic to $\mathfrak{sl}_2$. For every orthogonal factor with $d_i = 4$ and every place p of F_i , put an edge $e = (i, p)$ between the two ideals acting on that block. For every unitary factor with $d_i = 2$ and every place p , put an edge between its simple ideal and a new vertex $t_{i,p}$, used only for this edge. We call these new vertices *torus vertices*. Orthogonal dimension-three ideals are included as ordinary vertices, even when no edge meets them.

An edge sign ε_e will denote the scalar action on its quadratic block after the preceding reduction. To remove all these signs using the shared derived group, we seek signs s_v at the ordinary vertices, and sign 1 at each torus vertex, such that

$$\varepsilon_e = s_u s_v \quad (e \text{ with endpoints } u, v). \quad (6.6)$$

Indeed, a central sign in SL_2 acts on its standard module by that sign. The orthogonal dimension-four standard module is the tensor product of the two endpoint modules, while the unitary dimension-two

standard module has just the ordinary endpoint's derived action. The same center acts trivially on the orthogonal dimension-three standard module.

The necessary compatibility is that the edge signs multiply to one around every cycle and along every path between torus vertices. These relations are also sufficient: path products from a base vertex give the signs s_v , as proved in Lemma 6.10. The role of the auxiliary tensors is to establish precisely these relations for every normalized tensor automorphism.

We therefore use every simple cycle of Γ , including a cycle formed by two parallel edges, and every simple path with both endpoints torus vertices. There are finitely many. Fix one such path or cycle γ . All its edges have the same real field F and the same place p under Lemma 6.2. For each embedding $\tau : F \hookrightarrow \mathbb{C}$, choose a Galois transport γ_τ of the path or cycle from p to τ . It uses the same list of original factor indices, although its simple ideals have been transported. At the original place choose the identity transport, so that $\gamma_p = \gamma$.

Viewing each edge as an interval, define the *grid* to be the Cartesian product of these graphs,

$$\mathcal{G}_\gamma = \prod_{\tau: F \hookrightarrow \mathbb{C}} \gamma_\tau.$$

Its vertices are tuples of vertices, one at each place; a grid step changes one coordinate along an edge. A *box* is a product $\prod_\tau e_\tau$, where $e_\tau = (I(\tau), \tau)$ is one chosen edge of γ_τ . Its vertices choose one endpoint of each edge. A square face of a box varies the endpoints at two distinct places and freezes all other vertices. At any torus endpoint we may select either of its two character lines; include all such choices in the finite construction.

Lemma 6.6 (Realization of a box). *For each box there is an auxiliary abelian variety with weight-one realization $W_\square$ and an algebraic tensor*

$$\mathcal{P}_\square \in (\text{End } W_\square)^{\otimes 2}$$

as in Proposition 5.1. At a chosen complex embedding, $W_\square$ contains the tensor representation attached to every vertex of the box: the standard two-dimensional representation at an ordinary vertex and the selected character line at a torus vertex, tensored over the places of F .

Adjoining all these finitely many $W_\square$ to the original system does not change its Hodge Lie algebra. Vertex representations with the same choices, whether occurring in different boxes or in an original W_i , are identified by algebraic weight-one correspondences.

Proof. Choose a totally real Galois closure $F' \subset \mathbb{R}$ of the embedded field F . Every embedding $q : F \hookrightarrow \mathbb{C}$ then takes values in F' . For each such q , take the four-dimensional quadratic block of factor $I(q)$ and extend scalars to F' through q . In the unitary case this is the underlying quadratic space of its two-dimensional Hermitian block. Let V' be their orthogonal sum together with sufficiently many negative Tate directions.

At every real embedding of F' , exactly one of the chosen blocks has a period plane: composition with that embedding permutes the places q , and their active places agree under the common-field identification. All other blocks are negative definite. Thus V' has signature $(2, d' - 2)$ at every real embedding. Choose the number of negative directions so that d' is sufficiently large and $d' \equiv 5 \pmod{8}$. The Witt index over F' is at least two. Indeed the nonarchimedean anisotropic dimension bound and the Hasse principle first give a hyperbolic plane; the residual form still has positive and negative directions at every real place and, in sufficiently large dimension, gives a second hyperbolic plane.

Proposition 5.1 applies to

$$W_\square = \text{Res}_{F'/\mathbb{Q}} C_{F'}^+(V')$$

and supplies its tensor $\mathcal{P}_\square$. The spin decomposition at the chosen embedding gives every endpoint choice in the four-dimensional blocks. The extra negative Tate directions supply only multiplicities, since the Hodge group acts trivially on them. This proves the assertion about vertex representations.

Here is the rational group action underlying this construction. Write Q_i for the four-dimensional quadratic space over F of the edge's factor, and put $Q_{i,q} = Q_i \otimes_{F,q} F'$. Extension of scalars through q gives a rational group homomorphism

$$\mathrm{Res}_{F/\mathbb{Q}} \mathrm{SO}(Q_i) \longrightarrow \mathrm{Res}_{F'/\mathbb{Q}} \mathrm{SO}(Q_{i,q}).$$

Compose these maps with the factor actions of G_0 , take their orthogonal sum over q , and act trivially on the added Tate directions. This defines

$$\rho_\square : G_0 \longrightarrow \mathrm{Res}_{F'/\mathbb{Q}} \mathrm{SO}(V').$$

The fiber product of G_0 with the corresponding spin covers, using all the maps $\rho_\square$, is a rational group mapping onto G_0 with finite kernel. The original Hodge circle together with its auxiliary KS actions lies in this fiber product. Consequently the enlarged Hodge group projects onto G_0 with finite kernel and has the same Lie algebra. Matching vertex choices therefore have the same Lie representation in all their occurrences. Connectedness gives the same group representation, and the weight-one correspondence argument of Lemma 6.3 makes their identifications and multiplicity projections algebraic. $\square$

Fix all the varieties in Lemma 6.6, and from now on let G, C denote the Hodge and algebraic tensor groups of this entire finite collection. We identify the Lie algebra of G with that of G_0 . In particular this collection satisfies the hypothesis of the preceding normalization argument. The preceding lemmas apply to this final collection. For a grid vertex u , let R_u be the abstract tensor product of its endpoint representations. Identify it with a multiplicity copy in every occurrence by weight-one Hodge embeddings. Every intertwiner between such copies is algebraic by Lemma 6.6. Hence any $c \in C(\mathbb{C})$ induces the same operator c_u on R_u through all these identifications. We may therefore choose multiplicity copies in a box using its Clifford tensor decomposition without changing c_u .

6.5 The signs of a normalized tensor automorphism

Now choose $c \in C(\mathbb{C})$ and normalize it by Lemma 6.5, using the CM part of this fixed final collection. For an edge $e = (i, p)$ of Γ , let

$$\varepsilon_e \in \{1, -1\}$$

be its scalar action on the associated quadratic block of V_i . For a unitary edge this is a sign by (6.5) with $d_i = 2$, and standard and dual have the same sign.

At each four-dimensional block, a lift to $\mathrm{Spin}_4 = \mathrm{SL}_2 \times \mathrm{SL}_2$ acts on the two endpoint modules. The normalized trivial adjoint action makes each semisimple SL_2 lift central, so its endpoint scalar is a sign. In the orthogonal case the two signs multiply to ε_e . In the unitary case one endpoint is a torus: if $a_e \in \{1, -1\}$ is the ordinary endpoint sign and λ_e is the selected torus character scalar, then $a_e \lambda_e = \varepsilon_e$. Thus λ_e and its inverse, the opposite character scalar, are signs as well. These endpoint signs may at present depend on the edge. On a word in an original W_i , the action of c is the product of the endpoint signs selected at its places. An overall sign of the spin lift can be absorbed into one block.

These are the edge signs in (6.6). We will prove their cycle and torus-endpoint relations by transporting a sign from one grid step to every parallel step, independently of the frozen vertices.

6.6 Transporting a sign across the grid

We first isolate a linear-algebra observation that will justify the subspaces used in the geometric tensor calculation.

Lemma 6.7 (Supports of an invariant pairing). *Let A, B be finite-dimensional complex vector spaces, let $U \subset A$ and $V \subset B$, and let $t \in U \otimes V$ induce an isomorphism $U^\vee \rightarrow V$. If automorphisms a of A and b of B satisfy $(a \otimes b)t = t$, then $a(U) = U$ and $b(V) = V$. If moreover $a|_U = \varepsilon \text{id}$ for $\varepsilon \in \{1, -1\}$, then $b|_V = \varepsilon \text{id}$.*

Proof. The images of the contraction maps $B^\vee \rightarrow A$ and $A^\vee \rightarrow B$ defined by t are exactly U and V . Thus preservation of t preserves both subspaces. The last assertion follows from the perfect pairing, since $\varepsilon^{-1} = \varepsilon$. $\square$

Fix one of the paths or cycles used to construct a grid. For a step along its edge $e = (i, p)$ at place p , freeze the vertices at every other place. If the local endpoint spaces are X, Y and the tensor product of the frozen spaces is T , the two grid representations are $X \otimes T$ and $Y \otimes T$. Inside the Hom space between them, consider

$$\text{Hom}(X, Y) \otimes \text{id}_T. \quad (6.7)$$

The reverse direction uses $\text{Hom}(Y, X) \otimes \text{id}_T$. Neither subspace is being assumed algebraic on its own.

Lemma 6.8 (Propagation across a box). *For the normalized element c , the action on (6.7), in either direction, is the scalar ε_e , independently of the frozen vertices at the other places of the grid.*

Proof. First choose every frozen vertex at an endpoint of the transported edge of factor i . The corresponding tensor words occur in the original W_i . Its endpoint-sign description shows that the action on (6.7) is ε_e in both directions: the frozen signs cancel, and the ratio of the two endpoint signs is their product.

It remains to move one frozen vertex across an edge at a different place q . Choose a box containing these two steps, with the other places fixed. Denote its four vertex representations by

$$R_{00} = X \otimes Z \otimes T_0, \quad R_{10} = Y \otimes Z \otimes T_0, \quad R_{01} = X \otimes Z' \otimes T_0, \quad R_{11} = Y \otimes Z' \otimes T_0.$$

Project $\mathcal{P}_\square$ to

$$\text{Hom}(R_{00}, R_{10}) \otimes \text{Hom}(R_{11}, R_{01}). \quad (6.8)$$

The second p -direction has been reversed. This projection is algebraic, since its vertex projections are weight-one Hodge endomorphisms. Figure 1 records the two directions.

We verify that the tensor in (6.8) is a perfect pairing between the two local Hom subspaces. Only the p -block of V' contributes. Clifford multiplication by a vector in that four-dimensional block interchanges its two half-spin spaces and acts as the identity on spectator slots, up to fixed grading signs. In a graded Clifford tensor model these signs are the chirality operators on preceding even-dimensional slots; after choosing the spectators they are scalars. The volume element in the total odd-dimensional Clifford module likewise acts as a scalar. Choose multiplicity copies in this Clifford tensor decomposition and transport the projected tensor to the fixed tensor models R_u . Its spectator factors are then identities; the transported actions are the already fixed operators c_u .

For the two two-dimensional half-spin spaces X, Y , the vector representation is $X \otimes Y$, and Clifford multiplication gives all of $\text{Hom}(X, Y)$ and all of $\text{Hom}(Y, X)$. More explicitly, with the

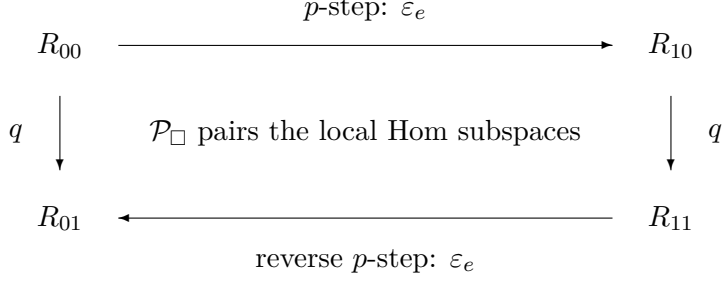


Figure 1: A square face of a box in the grid of transported paths or cycles. Only the distinct place coordinates p and q vary. The projected tensor pairs the local Hom subspace represented by the upper p -arrow with the one represented by the reversed lower arrow. Preservation of this perfect pairing transports the scalar sign when the frozen q -vertex changes. The vertical arrows indicate that change of vertex, not a chosen linear map.

skew-form identifications $j_X : X \rightarrow X^\vee$ and $j_Y : Y \rightarrow Y^\vee$, the two maps belonging to $x \otimes y$ are, up to nonzero scalars,

$$y \otimes j_X(x), \quad x \otimes j_Y(y).$$

The inverse quadratic metric therefore gives a nondegenerate tensor between the two Hom spaces, with identities on the frozen factors. If an endpoint is a torus character line in Y , project to that line. The first Hom space consists of maps from X into the selected line, and the second of maps from that same line to X . In the displayed formulas these use respectively the selected character component of y and its opposite under j_Y . They remain perfectly paired by the metric. Thus the projected tensor has exactly the local Hom subspaces as its minimal tensor supports.

Since c preserves $\mathcal{P}_{\square}$ and all the vertex projections, Lemma 6.7 first makes these Hom subspaces invariant and then transfers the sign ε_e from the upper one to the lower one. Reversing the roles of the two directions gives the reverse propagation as well. Moving the frozen places successively along their paths proves the assertion for every vertex choice. $\square$

Lemma 6.9 (Cycle and endpoint relations). *The edge signs of Γ satisfy*

$$\prod_{e \in \gamma} \varepsilon_e = 1 \tag{6.9}$$

for every cycle γ and for every path γ whose two endpoints are torus vertices.

Proof. It suffices to use simple cycles and simple paths, since cancellation of repeated edges reduces the general case to them. Fix one such path or cycle and a place p . Freeze all other places at the starts of their transported paths. At the initial grid vertex, the action of c is scalar: all these endpoint choices occur in the original representation belonging to the first edge.

Follow the path at place p . Suppose c is scalar a on its current vertex representation. By Lemma 6.8, conjugation on the local Hom space for the next edge e is ε_e . Hence

$$c_{\text{next}} f = \varepsilon_e a f \quad \text{for every } f \in \text{Hom}(X, Y) \otimes \text{id}_T.$$

The images of these maps span the next vertex representation. Thus its scalar is $\varepsilon_e a$. Around a cycle one returns to the same representation and obtains (6.9). For a torus-to-torus path, both endpoint words consist only of character slots, including at the frozen places. They belong to W_{CM} , so their scalars are both one. The same calculation proves the endpoint relation. $\square$

Lemma 6.10 (Vertex signs). *There are signs s_v at the vertices of Γ , with $s_v = 1$ at every torus vertex, such that*

$$\varepsilon_e = s_u s_v \quad \text{for every edge } e \text{ with endpoints } u, v.$$

Proof. In each connected component choose a base vertex, choosing a torus vertex when one is present, and give it sign one. Define the sign of any other vertex as the product of the edge signs along a path from the base. The cycle relations in Lemma 6.9 make this independent of the path. If another torus vertex occurs, the endpoint-path relation gives it sign one. The edge formula follows immediately. Isolated vertices may also be assigned sign one. $\square$

6.7 Completion of the mixed-product argument

Proof of Theorem 1.1. Take $c \in C(\mathbb{C})$ for the final collection fixed after Lemma 6.6. Lemma 6.5 removes an element of $G(\mathbb{C})$ so that the only remaining actions on the V_i are the edge signs of Γ . By Lemma 6.10, these are products of signs assigned to the shared $\mathfrak{sl}_2$ ideals, with sign one at the torus vertices. Use those signs in the centers of the corresponding simply connected derived factors. On an orthogonal dimension-four block their action is the product of the two endpoint signs; on a unitary dimension-two standard block it is the ordinary-vertex sign. Their action on orthogonal dimension-three blocks is trivial. They therefore remove every remaining action on the V_i . This proves (6.1) over $\mathbb{C}$, and hence over $\mathbb{Q}$.

Every rational Hodge tensor in the V_i is now fixed by C and is therefore algebraic. Apply the algebraic Künneth projectors to each factor of $S_1 \times \cdots \times S_m$. Besides the transcendental summands in degree two, the pieces are generated by the unit, divisors, and the point class, all algebraic. Thus every Hodge class in a Künneth summand is a sum of the algebraic transcendental tensors just obtained tensored with these elementary classes. Products and the projectors realize them in the required codimension. This proves surjectivity of the rational cycle-class map for every ordered list, including repetitions, and every codimension in the statement. $\square$

Part II

The realization theorem

The preceding argument reduces Theorem 1.1 to Proposition 2.1. We now prove that proposition. Its two conclusions, prescribed Euler pairings and a bound on Ext^2 , will come from one Lagrangian immersion: its fundamental class controls the pairings, while its second cohomology controls the extension space. The construction must also remove the disk obstruction and identify the category generated by the embedded mirror tests. These are the successive tasks of this part.

7 A Lagrangian immersion with controlled cohomology

We begin the proof of Proposition 2.1 by constructing a geometric representative of its input class a . Two features of that representative will control its Floer theory: ambient classes detect its degree-two cohomology, and its self-intersections contribute only odd-degree generators. We establish these features together. All homology and cohomology in this section have rational coefficients unless otherwise indicated.

Proposition 7.1. *Let $D = Y \times T_g$, ω , and n be as in Section 2, with g even and $n \geq 12$. Suppose that $a \in H^n(D, \mathbb{Q})$ satisfies $\omega a = 0$, and that its annihilator*

$$\mathcal{Z} = \{x \in H^2(D, \mathbb{Q}) : xa = 0\}$$

satisfies the injectivity hypothesis (2.4). There exist a closed connected spin n -manifold L , a positive integer ν , and a Maslov-zero Lagrangian immersion $i : L \rightarrow D$ such that

$$\text{PD } i_*[L] = \nu a, \tag{7.1}$$

$$i^* : H^2(D, \mathbb{Q})/\mathcal{Z} \xrightarrow{\sim} H^2(L, \mathbb{Q}), \tag{7.2}$$

and

$$i_* : H_{n-2}(L, \mathbb{Q}) \longrightarrow H_{n-2}(D, \mathbb{Q}) \quad \text{is injective.} \tag{7.3}$$

The immersion is surjective on fundamental groups and has only transverse double points, with no triple points. With an integer grading whose orientation agrees with the orientation of L , both ordered branch indices at every double point are odd.

We first realize the class with the required tangential structure and low cohomology. The immersion theorem then supplies a graded Lagrangian representative. To control the double-point signs, we place all of its crossings inside a contractible exact domain and apply the flexibility results of Eliashberg–Murphy and Ekholm–Eliashberg–Murphy–Smith. This adapts the local-domain construction of [Ope26c, Section 3]; the choice of filling classes below also treats the nonzero π_2 of the K3 factor.

7.1 Tangential realization and low-dimensional surgery

We seek a representative whose degree-two cohomology is exactly $H^2(D, \mathbb{Q})/\mathcal{Z}$ and whose complexified tangent bundle is identified with the ambient tangent bundle. We impose these two conditions by lifting the representing map: a torus bundle kills the classes in $\mathcal{Z}$, and a spin pullback records the required stable tangent identification. The following calculation shows that the lift introduces no further degree-two classes; surgery will then transfer this degree-two description to the representing manifold.

Choose independent integral classes $z_1, \dots, z_r$ spanning $\mathcal{Z}$ over $\mathbb{Q}$, and let $\pi_0 : \mathcal{T}_0 \rightarrow D$ be the principal r -torus bundle with these Chern classes. Choose the natural stable SU structure on TD , and form the homotopy pullback

$$\mathcal{T} = \mathcal{T}_0 \times_{B\text{SU}}^h B\text{Spin}, \quad \pi : \mathcal{T} \longrightarrow D,$$

where $B\text{Spin} \rightarrow B\text{SU}$ is complexification. The tautological stable real bundle ξ on $\mathcal{T}$ has virtual rank n , is spin, and comes with a stable identification

$$\xi \otimes \mathbb{C} \simeq \pi^*TD \tag{7.4}$$

that retains the determinant trivializations.

Lemma 7.2. *Pullback induces*

$$H^2(\mathcal{T}, \mathbb{Q}) \simeq H^2(D, \mathbb{Q})/\mathcal{Z}.$$

Moreover, $\text{PD}^{-1}a \in H_n(D, \mathbb{Q})$ is in the image of $\pi_ : H_n(\mathcal{T}, \mathbb{Q}) \rightarrow H_n(D, \mathbb{Q})$.*

Proof. The compact Kähler manifold D is formal over $\mathbb{R}$ [DGMS75]. We first compute over $\mathbb{R}$. A real model for its torus bundle is obtained from $H^*(D, \mathbb{R})$ by adjoining degree-one generators $u_1, \dots, u_r$ with $du_j = z_j$. In degree two, the horizontal classes give $H^2(D, \mathbb{R})/(\mathcal{Z} \otimes \mathbb{R})$. The differential on $H^1(D, \mathbb{R}) \otimes \langle u_1, \dots, u_r \rangle$ is the scalar extension of the map in (2.4), and is injective. The differential on $\wedge^2 \langle u_1, \dots, u_r \rangle$ is also injective:

$$d(u_j u_l) = z_j u_l - z_l u_j,$$

and the z_j are independent. The images of these two differentials have different vertical degrees, so cannot cancel.

The fiber of $B\text{Spin} \rightarrow BSU$ is two-connected. Over $\mathbb{R}$, complexification takes each even-index Chern generator to a nonzero multiple of a free Pontryagin generator, and every odd-index Chern generator to zero. In the relative Koszul model for the second pullback, eliminate the even-index pairs. What remains is the torus bundle model with generators of degrees $4j + 1$, $j \geq 1$, whose differentials are the odd Chern classes of TD . These classes vanish: TD is the sum of the tangent bundle of a K3 surface, when present, and a trivial bundle. In particular the second pullback changes neither the preceding degree-two calculation nor its pullback map.

Define a functional on the degree- n part of this model by pairing a horizontal class $v \in H^n(D, \mathbb{R})$ with a , and setting the functional equal to zero on terms with positive vertical degree. It is a cycle in the dual complex. Indeed, a differential with a horizontal term is a sum of multiples of the z_j , and $az_j = 0$; the remaining new generators have zero differential. Its induced real homology class maps to $\text{PD}^{-1} a$. The cohomology pullback and homology pushforward are defined over $\mathbb{Q}$. Faithfulness of scalar extension from $\mathbb{Q}$ to $\mathbb{R}$ therefore descends the asserted kernel and isomorphism, as well as membership of the rational class $\text{PD}^{-1} a$ in the image of pushforward. $\square$

Lemma 7.3. *There are a closed connected spin n -manifold L_0 , a map $f : L_0 \rightarrow D$, and a positive integer ν satisfying (7.1)–(7.3) with f in place of i . They admit a stable identification*

$$TL_0 \otimes \mathbb{C} \simeq f^*TD \tag{7.5}$$

with the specified determinant homotopy. The map is surjective on fundamental groups, and L_0 has two additional free circle generators mapping trivially to D . After fixing ν , its Euler characteristic can be made arbitrarily large while preserving these assertions.

Proof. Use the spin bundle ξ on $\mathcal{T}$, including (7.4), as a stable tangential structure. With our virtual-rank- n convention, the group of n -dimensional bordism classes carrying this structure is $\pi_0 \text{Th}(-\xi)$ by the Pontryagin–Thom construction; see [Tho54] and [RF24, Section 2.3]. Rational stable homotopy and the Thom isomorphism give

$$\pi_0 \text{Th}(-\xi) \otimes \mathbb{Q} \simeq H_0(\text{Th}(-\xi); \mathbb{Q}) \simeq H_n(\mathcal{T}; \mathbb{Q}).$$

Thus a positive integral multiple of the lift in Lemma 7.2 is represented by a closed manifold mapping to $\mathcal{T}$, with its stable tangent bundle identified with the pullback of ξ . Composing with π gives (7.5) and the required fundamental class. We now modify the representative by bordisms over $\mathcal{T}$.

Put $\Gamma = \pi_1(\mathcal{T})$. The torus-bundle homotopy sequence exhibits its fundamental group as an extension of $\pi_1(D)$ by a quotient of $\mathbb{Z}^r$; the second pullback has two-connected fiber. Consequently Γ is polycyclic, and $\Gamma \rightarrow \pi_1(D)$ is surjective. The space $\mathcal{T}$ has a CW model with finite skeleta in the degrees needed here: the base and torus bundle are compact, and the characteristic-class

pullback uses classifying spaces and loop spaces of finite type. The group ring $\mathbb{Z}\Gamma$ is Noetherian [Hal54].

Join the components of the representative by one-handles and add loops mapping to generators of Γ . Its fundamental-group map is then surjective. Its kernel is normally finitely generated, since both groups are finitely presented. Surgery on disjoint embedded representatives of normal generators makes that map an isomorphism. Each surgery uses a null homotopy over $\mathcal{T}$; a twist of the normal framing on a circle makes the stable tangential data extend.

Next make the map on π_2 surjective by adding $S^2 \times S^{n-2}$ summands mapped through S^2 , for a finite set of module generators of the missing classes. Such a pulled-back spin bundle is stably trivial over S^2 , so these additions carry the prescribed tangential data. The kernel on π_2 is finitely generated as a $\mathbb{Z}\Gamma$ -module: this follows from the finite cellular chain modules of the universal covers and Noetherianity. Represent its generators by disjoint embedded two-spheres and perform surgery on them, using their null homotopies over $\mathcal{T}$. Their normal bundles are trivial: the spin condition makes their stable normal bundles trivial, and the rank is in the stable range. The stable framing extends over surgery because $\pi_2 \text{SO} = 0$. Since $n \geq 12$, the embedding dimensions and the reverse-handle indices preserve the already established fundamental-group assertion. These operations make π_2 an isomorphism as well. The resulting map to $\mathcal{T}$ consequently induces an isomorphism on H^2 .

Now take connected sums with two copies of $S^1 \times S^{n-1}$, and with as many copies of $S^4 \times S^{n-4}$ as desired, mapping these summands constantly to $\mathcal{T}$. These products are stably parallelizable and preserve the tangential structure and pushed fundamental class. They do not change H^2 . The first two summands provide the extra null-image circles; each subsequent summand raises the Euler characteristic by two. Lemma 7.2 therefore proves (7.2).

To verify (7.3), write any $u \in H_{n-2}(L_0, \mathbb{Q})$ as $f^*x \cap [L_0]$ using Poincaré duality and the surjectivity in (7.2). The projection formula gives

$$f_*u = x \cap f_*[L_0] = \nu \text{PD}^{-1}(xa).$$

If this is zero, then $x \in \mathcal{Z}$ and $f^*x = 0$, so $u = 0$. □

Since $\omega \in \mathcal{Z}$, the map just constructed satisfies $f^*[\omega] = 0$. Destabilize (7.5) in the unitary stable range over the n -dimensional manifold L_0 , retaining the determinant homotopy. Restriction to the real tangent bundle gives a formal Lagrangian immersion of Maslov class zero. The Gromov–Lees theorem realizes it by a Lagrangian immersion homotopic to f [Gro86, RF24]. We continue to call this immersion f , and choose its integer grading with the orientation of L_0 .

7.2 Chosen fillings of zero area

The remaining task is to alter the double points without changing the cohomological properties. To do this inside a contractible exact domain, the loops formed by the two branches of each crossing must admit chosen fillings of zero symplectic area. No uniqueness of a filling area is asserted or required.

Lemma 7.4. *After Lagrangian perturbations, f is self-transverse without triple points, and there are pairwise disjoint embedded arcs in L_0 joining the two preimages of each double point such that every image loop has a chosen filling in D of symplectic area zero. The perturbations preserve the homotopy class and integer grading.*

Proof. Choose fillings for the two extra null-image circles from Lemma 7.3. In an immersed cotangent neighborhood, replacing the zero section by a small closed one-form changes each chosen filling area

by the corresponding period, up to a fixed sign. The two periods can be adjusted independently. Choose the resulting areas to generate a dense subgroup of $\mathbb{R}$. Concatenations and reversals of these circles are given the concatenated and reversed fillings, so their chosen areas add. Small exact perturbations now make the immersion self-transverse and remove triple points, without changing these periods.

For a double point $f(a_j) = f(b_j)$, begin with a path from a_j to b_j . Surjectivity on fundamental groups allows insertion of a source loop that makes its image contractible. Choose a filling. By inserting combinations of the two null-image circles with their chosen fillings, make its area arbitrarily small. Choose disjoint source patches at the a_j , avoiding every other double-point preimage. An exact perturbation by dF_j , where F_j is constant near a_j and zero off its patch, changes this path's filling area by $F_j(b_j) - F_j(a_j)$, up to the same sign convention. It fixes the sheet near the crossing. Thus it makes the area exactly zero. The endpoint formula remains valid if the path visits the patch several times. With the patches fixed, a uniform sufficiently small bound on these perturbations preserves all crossings and creates no new ones; take the preceding residual areas within that bound.

Finally, homotopies of the source paths relative to their endpoints preserve the chosen areas, because $f^*\omega = 0$ as a differential form. General position makes the finitely many paths embedded, mutually disjoint, and disjoint from the other double-point preimages. Grading and spin data transport through the Lagrangian perturbations. $\square$

7.3 A contractible exact neighborhood

Lemma 7.5. *There is a compact contractible Weinstein domain $(P, \lambda) \subset (D, \omega)$, with $d\lambda = \omega|_P$, such that $W = f^{-1}(P)$ is a disk containing all double-point preimages. Its primitive satisfies $f^*\lambda|_W = 0$, and its Liouville field is tangent to the immersed sheets. The immersion is cylindrical along ∂W . At one interior point, W has a single linear sheet in a radial Darboux ball.*

Proof. The source and target neighborhoods have different topology. A tree joining the double-point preimages has a disk as its source thickening, but its image acquires a loop at each double point. We first thicken that image to a Weinstein zero- and one-handlebody P_1 . We then attach two-handles along its free generators, with cores disjoint from the immersion. The target becomes contractible while its preimage remains a disk. We give the primitive, collar, and filling details needed for this construction, adapting [Ope26c, Lemma 3.5].

Thickening the tree and its image. Extend the arcs of Lemma 7.4 to an embedded tree $\Theta \subset L_0$ by connector arcs avoiding all other sheet intersections. Its image $G = f(\Theta)$ has precisely the identifications at the double-point pairs. Each chosen source arc becomes one of the free graph cycles, with its previously chosen zero-area filling. Subdivide an ordinary edge to obtain a vertex with only one preimage. Figure 2 illustrates the resulting spines.

At each vertex take a small radial Darboux ball. The single sheet, or the two transverse sheets at a double point, can be straightened simultaneously. For two sheets, first make one horizontal; the other is a gradient graph over the vertical plane, and a symplectic shear straightens it while fixing the first. Arrange the incident tree edges as distinct radial rays. Thin Weinstein one-handles along the remaining image arcs join these balls and carry strips in the corresponding source sheets.

Here are compatible models for these attachments and the two-handles used below. In symplectic coordinates (u, v, z) , with $u, v \in \mathbb{R}^j$ and $z \in \mathbb{R}^{2(n-j)}$, use the Liouville field

$$Z_j = (-u, 2v, z/2), \quad j = 1, 2. \quad (7.6)$$

A box $|u| \leq 1$, $|(v, z)| \leq \varepsilon$ has negative face near $|u| = 1$, isotropic core $v = z = 0$, and a quadratic Lyapunov function with one critical point of index j . For $j = 1$, the subspace $v = 0$ with z in a fixed Lagrangian plane is an invariant Lagrangian strip.

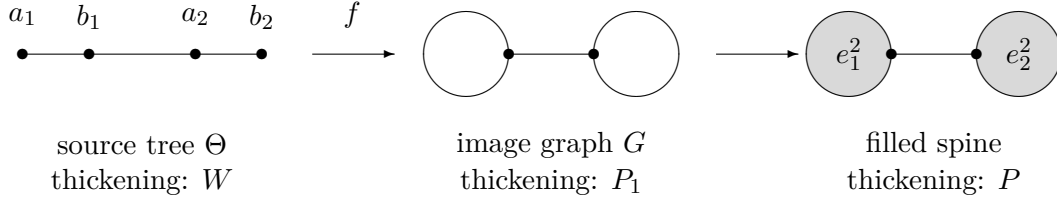


Figure 2: Schematic spines for two double points. Identifying a_j with b_j turns a contractible source tree into a graph with loops. Two-handles kill these loops in the target; their filling interiors miss the immersion, so the preimage remains the disk W . The picture represents the abstract spines, not an embedding in D .

The attaching contact identifications can be chosen strict for the specified contact forms. Indeed, near an isotropic attaching locus choose a slice tangent to the contact plane along that locus. The Reeb flow writes the contact form as $dt + \beta$ on a neighborhood of the slice. Match the slices by the symplectic isotropic neighborhood theorem with the prescribed tangent and symplectic normal data. The resulting difference of primitives is a closed form that restricts to zero on the attaching locus. It is dh , with $h = 0$ there, since the neighborhood retracts onto the locus. Adjusting the Reeb coordinate by h makes the contact identification strict. The same argument applies to the Legendrian patches in the sheets. Extending by Liouville flow gives collars preserving the primitives.

For the one-handles, choose transverse strip framings along the source arcs and extend the prescribed end germs over each strip. The relative Lagrangian neighborhood theorem extends these maps symplectically with the collar germs fixed: after fixing tangent data, the remaining linear lifts are contractible, and the tubular construction and Moser argument are relative to the collars. Shrinking the handles keeps them embedded, mutually disjoint away from their attachments, and disjoint from every other sheet. The model primitives glue through the strict collars. After rounding the positive corners, we obtain a Weinstein zero- and one-handlebody P_1 with spine G . Its preimage is a thickening of the source tree, hence a disk, and its primitive vanishes on the sheets. The singled-out radial ball can be retained inside P_1 .

Isotropic fillings outside the one-handlebody. The loops of P_1 now have to be killed without meeting any source sheet. The fillings must retain their chosen zero-area classes: contractibility of the attaching circles alone would not control area in the presence of $\pi_2(D)$. Choose a free basis of $\pi_1(P_1)$ represented by disjoint isotropic circles in ∂P_1 , avoiding the Legendrian boundary of the source disk. These representatives exist by the relative subcritical contact isotropic embedding theorem [EM02, Theorem 12.4.1]; formal tangent lines can be moved into the contact distribution, and all the avoidance conditions are in the general-position range.

Each attaching circle inherits a zero-area filling in D : join it in P_1 to its graph cycle by an annulus and append the chosen filling. Both boundary primitive integrals vanish, so the annulus has area zero. We may represent this same filling class outside P_1 . In fact, recovering a $2n$ -dimensional zero- and one-handlebody from its boundary uses only handles of indices $2n$ and $2n - 1$. The inclusion of the exterior into D therefore has no relative homotopy in the dimensions of this disk and its deformation. This preserves the chosen relative disk class, not just the contractibility of the attaching circle. Consequently its symplectic area remains zero even when $\pi_2(D) \neq 0$. Prescribe the initial collar to be the outward Liouville cylinder. General position relative to these collars gives embedded filling disks with disjoint interiors, all missing $f(L_0)$.

We next make the disks isotropic while retaining those collars. Use a prequantization circle bundle for the integral form ω . The collars lift horizontally. In a trivialization over a filling disk, the zero area gives winding number zero for its boundary lift, so this lift extends over the disk. The space of isotropic ordered two-frames in the contact distribution retracts to $U(n)/U(n-2)$ and is simply connected. Thus the prescribed isotropic derivative on the collar extends as a formal isotropic monomorphism over the disk. The ordinary real two-frame space in rank $2n+1$ is two-connected, so the derivative of the lifted disk is homotopic to that monomorphism relative to the collar. The relative, C^0 -dense subcritical contact embedding h -principle realizes these data, fixing a slightly smaller collar.

Projection gives isotropic immersions of the disks into D . They can be chosen embedded and mutually disjoint: contact Hamiltonian variations near distinct lift points give arbitrary independent projected variations, and $2+2 < 2n$. The same argument preserves avoidance of the immersed source because $2+n < 2n$. Smallness protects avoidance away from the fixed collars; the collars themselves protect the attachments.

Attaching the two-handles. We now have embedded isotropic disks with disjoint interiors outside P_1 , prescribed outward collars, and no intersections with the immersion. Choose symplectic normal framings over these disks. They restrict to the contact symplectic normal framings needed at the attaching circles. Parameterize the attaching circles by parameterizations extending over the disks. The relative isotropic neighborhood theorem and relative Moser argument now embed thin index-two handles with the prescribed strict collars and model (7.6). These handles miss the immersion. Their primitives glue, and rounding corners produces P .

Its spine is the graph G with a two-cell attached along each free generator. After collapsing a maximal tree, these attaching maps form a free basis of a bouquet of circles; the resulting complex is contractible. Thus P is contractible. The two-handles have not changed the preimage disk, its primitive-zero sheets, or its outer cylindrical collar. The radial single-sheet ball also remains intact. $\square$

We have now localized all double points in an exact simply connected target. This is what permits their algebraic signed count, rather than a fundamental-group-valued intersection invariant, to control the final replacement.

7.4 Double-point signs and the integer grading

Proof of Proposition 7.1. Write $n = 2k$. For the immersion supplied by the preceding lemmas, the self-intersection formula reads

$$2I(f) = f_*[L_0] \cdot f_*[L_0] - (-1)^k \chi(L_0). \quad (7.7)$$

Here the normal Euler number is $(-1)^k \chi(L_0)$, since the normal bundle of an oriented Lagrangian is its tangent bundle with this orientation factor. In Lemma 7.3, choose the Euler characteristic large enough that $I(f)$ is nonzero and has sign $(-1)^{k-1}$. Put $d = |I(f)|$.

Complete the domain P of Lemma 7.5 by its positive cylindrical end, extend W by its cylinder, and remove the center of its single-sheet radial ball. The punctured ball is a negative Liouville end. The source is now $S^{n-1} \times \mathbb{R}$, with the standard real-plane Legendrian sphere ϕ_0 at that end. The target is simply connected: P was contractible, and removing a point in dimension $2n$ does not change its fundamental group.

The stabilization traces of [EEMS13, Lemma 3.2 and proof of Lemma 3.4] have signs

$$\phi \longrightarrow \phi_1 : (-1)^{k-1}, \quad \phi_1 \longrightarrow \phi : (-1)^k, \quad (7.8)$$

where ϕ_1 is a loose stabilization and the arrows run outward along the symplectization. Insert d reverse traces in separated collars near the puncture, from the d -fold stabilization ϕ_d at the negative end to ϕ_0 outward. The new negative link is connected and loose, and its total signed double count is zero by (7.7).

Eliashberg–Murphy’s exact cancellation theorem [EM13, Theorem 2.3] now applies. The target is a simply connected Liouville manifold with a negative end; the immersion is proper, cylindrical there, self-transverse, and has finitely many double points. It is exact because $H^1(S^{n-1} \times \mathbb{R}; \mathbb{R}) = 0$. The loose component and vanishing signed count were just arranged. The theorem’s extra width condition concerns source dimension three, so is irrelevant here. We obtain a compactly supported Hamiltonian regular homotopy to an embedding, leaving both cylindrical tails fixed.

In the unchanged negative tail insert d forward traces from ϕ_0 to ϕ_d , and fill the center by the real plane. The resulting disk immersion has exactly d double points, all of sign $(-1)^{k-1}$. Its noncylindrical part is compact. A sufficiently long negative Liouville flow in the filled positive completion brings that part inside P . Backward flow exists for all time; points of P cannot leave it backward, while the positive end is translated toward P . The flow rescales the symplectic form and hence preserves Lagrangianity. The cylindrical set is unchanged, so its intersection with the original outer collar is unchanged. Trim at that collar and glue to the immersion outside W .

This replaces a source disk by a source disk with the same collar. A collar identification extends continuously over the disks, and their maps into the contractible P are homotopic relative to the collar. It follows that the pushed fundamental class and the maps on degree-two and degree- $(n-2)$ (co)homology are unchanged, under the identifications through the common complement. The same complement carries every first homology class and the fundamental-group map. The boundary sphere has a unique spin structure, so spin extends over the replacement disk. The Maslov class remains zero. Choose the extended integer grading with the original orientation parity.

For graded oriented Lagrangian intersections in real source dimension $2k$, a generator of degree $|x|$ has intersection sign

$$(-1)^k (-1)^{|x|}.$$

The sign $(-1)^{k-1}$ obtained above therefore makes every branch degree odd. Reversing the ordering replaces its degree by $n - |x|$, which is also odd because n is even. The disk replacement has only transverse double points and introduces no triples or intersections with the unmodified source. Together with Lemma 7.3, this proves all assertions. $\square$

8 Removing the disk obstruction

Let $i : L \rightarrow D$ be the graded spin immersion of Proposition 7.1. Its transverse self-intersection generators have odd integer degrees, and

$$H^2(L, \mathbb{Q}) \simeq H^2(D, \mathbb{Q})/\mathcal{Z}, \quad i_* : H_{n-2}(L, \mathbb{Q}) \longrightarrow H_{n-2}(D, \mathbb{Q}) \text{ is injective.} \quad (8.1)$$

We will use the second assertion to remove the curvature of its Fukaya algebra. The first will then bound its degree-two endomorphisms. The geometric ingredient is a Stokes identity for disks with one interior marked point and no external boundary marked point. We construct that identity geometrically, then use filtered transfer to remove curvature.

8.1 Branes, gradings, and the filtered construction

Choose countably many Hamiltonian copies of the product test branes, ordered by the positive integers, with each test occurring cofinally. Use product Hamiltonian perturbations, small enough

that the torus factors remain perturbed graphs. Arrange pairwise transversality and absence of triple intersections, including intersections with L . When both factors are present, impose these conditions separately on the factors of all embedded copies. Hamiltonian variations supply the required tangent directions, so these are countably many generic conditions.

Fix a split compatible almost complex structure J_0 with no nonconstant spheres and no nonconstant disks on any embedded test copy. The sphere and disk exclusion on the quartic factor is the genericity assertion used in the test-brane construction; on the torus factor it follows from the vanishing of the relevant second relative homotopy groups. No disk exclusion is imposed on L . All category constructions in this section involve a finite subcollection of the copies together with L .

Write Λ_0 for the universal Novikov ring with complex coefficients and Λ for its fraction field; its monomials are written T^λ , with valuation λ . We use the immersed cyclic de Rham construction of Abouzaid–Fukaya–Oh–Ohta–Ono [AFO⁺26, Theorems 9.3 and 9.6]. It gives a curved cyclic filtered A_∞ category with homotopy units for finite collections of closed spin immersions in a compact symplectic manifold. Here the relative spin background and bulk class are zero. The morphism spaces are forms on the source diagonals and the generators at transverse intersections. We denote the operations by m_k , their cyclic pairing by $\langle \cdot, \cdot \rangle$, and the curvature by m_0 .

These operations have integer degree $2 - k$. Indeed, the brane gradings specify integer indices for the cap paths at every transverse end, including self-intersection ends. After capping, the boundary grading has Maslov index zero. There is no ambiguity from spheres because the ambient first Chern class vanishes. The same index calculation applies to parameterized operations. Energy monomials have degree zero, and all contractions, transfers, and comparison maps below are chosen homogeneous. Thus the grading is retained by the construction itself, including the shifted degrees or auxiliary variables used in a particular model.

We use Kuranishi structures with normalized boundaries and corners, outer collars where required, and continuous-family perturbations (CF perturbations). The local chart theorem and its evaluation submersion criterion are [FOOO18, Theorem 7.1, Condition 7.11, and Lemma 7.12]; compatible systems arise from disk-component-wise obstruction data [FOOO24a, Theorems 5.3 and 5.4]. We use the corrected chart conventions of [FOOO24b]. The analytic gluing input is the exponential control of the gluing maps and their length derivatives [FOOO24c, Assumption 3.12 and Theorems 3.13, 6.4, 8.16, and 8.17]. For transverse immersed ends, the fixed-jump charts and spin determinant cappings are those of [AJ10, Sections 4–5], as incorporated in the immersed construction of [AFO⁺26].

In particular, the nodal chart uses component surjectivity augmented by matching equations at ordinary boundary nodes and by the transverse asymptotic conditions at jumps. Auxiliary markings stabilize the domains, and the obstruction spaces are transported on compact cores before gluing. These are the local inputs needed below when the total curve has no external boundary mark. All the additional spaces used for that purpose have one interior mark and no sphere components; there is no gluing at an angular interior node.

The construction is first made through finite energy and arity cutoffs. Filtered pseudo-isotopies, homotopy limits, and canonical models then give the complete category [AFO⁺26, Sections 6, 9–10, and 12, in particular Propositions 9.55 and 12.10]. We keep J_0 fixed in this passage. The following observation gives a common energy filtration for a finite list of branes.

Lemma 8.1 (Energy values). *For a finite list of the immersed and embedded branes above, the energies of J_0 -holomorphic polygons form a locally finite subset of $\mathbb{R}_{\geq 0}$. In particular, the nonconstant polygons have a common positive energy lower bound, independent of their number of marked points.*

Proof. Let $K \subset D$ be the union of their images. It has the homotopy type of a finite complex and has arbitrarily small regular neighborhoods: away from the finitely many double points these are

ordinary tubular neighborhoods, and at a double point they are the standard transverse-double neighborhoods. Relative de Rham theory therefore represents every class in $H^2(D, K; \mathbb{R})$ by a smooth closed form that vanishes near K .

The symplectic form has a relative lift that computes the polygon areas. To see this directly, construct near K a primitive α of ω whose pullback to every sheet is zero. At a double point, simultaneous straightening of the transverse Lagrangians gives a radial primitive with this property. On the remaining pieces use the Lagrangian neighborhood theorem. After shrinking the neighborhoods, their overlaps at the ends of the pieces are collars on disjoint sheets. The differences of the primitives are exact there, with potentials zero on the sheets, by the relative Poincaré lemma. These potentials patch the primitives. Cutting off α gives a closed form $\tilde{\omega}$ vanishing near K and representing the relative lift. Stokes' theorem, with the usual decay at jump ends, gives

$$\int u^* \tilde{\omega} = \int u^* \omega$$

for every polygon u .

Choose closed forms vanishing near K that represent a basis of $H^2(D, K; \mathbb{R})$. Their integrals over a J_0 -holomorphic polygon are bounded in absolute value by fixed constants times its area: this is the pointwise comass estimate for the compatible metric. A polygon defines an integral class in $H_2(D, K; \mathbb{Z})$, whose quotient by torsion is a lattice. The preceding bounds leave only finitely many lattice points below a fixed area cutoff. Pairing with $[\tilde{\omega}]$ leaves only finitely many energies. Energy positivity now gives the asserted lower bound whenever a nonconstant polygon exists. If none exists, any positive lower bound will do. $\square$

The additive monoid generated by these energies is locally finite as well: below an energy bound B , each summand belongs to a finite set and a sum has at most B/δ positive summands, where $\delta > 0$ is the gap. This is the filtration used for the transfer operations.

At energy zero, the unary operation is the de Rham differential on the source diagonals and zero on transverse-point generators; the constant strip with no stabilizing mark is unstable. Transfer at energy zero consequently gives the graded canonical space

$$C_L^{\text{can}} = H^*(L; \mathbb{C}) \otimes \Lambda_0 \oplus \bigoplus_{x \text{ a branch generator}} \Lambda_0 x. \quad (8.2)$$

The diagonal transfer can be made cyclic using a compatible de Rham contraction. Homotopy units will give cohomological units after curvature is removed. Since every branch generator has odd degree, the degree-two part of this canonical space is exactly $H^2(L; \mathbb{C}) \otimes \Lambda_0$. Removing curvature will therefore give the cohomological bound in the following proposition, which is the output of this section.

Proposition 8.2. *For every finite subcollection of the embedded test copies, L and those copies define an uncurved proper cohomologically unital cyclic A_∞ category over Λ . Its split-closed triangulated hull has Serre functor $[n]$, and*

$$\dim_\Lambda \text{Hom}^2(L, L) \leq b_2(L) = b_2(D) - \dim_{\mathbb{Q}} \mathcal{Z}. \quad (8.3)$$

For a sequence of embedded copies with strictly increasing labels, the operations are the geometric disk operations at J_0 , up to filtered pseudo-isotopy through every finite cutoff.

The proof occupies the rest of the section. To remove curvature, we construct scalar operations on disks with an interior root, relative to the ordinary cyclic disk operations. Their Stokes identity detects the leading curvature class against ambient cohomology and remains valid after positive-valuation boundary insertions. Filtered transfer then gives the proposition.

8.2 A Stokes identity with one interior mark

Fix the collared componentwise Kuranishi data and CF-perturbations defining the cyclic boundary-rooted disk operations through a finite energy and arity cutoff, enlarged to contain the factors of all boundary strata that will occur. We construct additional scalar operations relative to these geometric choices. Their disks carry one distinguished interior mark and cyclically ordered boundary inputs, but no boundary output. The component containing the interior mark roots a broken disk tree. Every other component has its ordinary boundary output at the node pointing toward that root.

Let η be a closed smooth $(n-2)$ -form on D . Pulling back η at the interior mark and the inputs at the boundary marks, and integrating, gives functionals

$$p_k^\eta(x_1, \dots, x_k) = \sum_{\beta} T^{\omega(\beta)} p_{k,\beta}^\eta(x_1, \dots, x_k), \quad k \geq 1. \quad (8.4)$$

We include all words of diagonal and transverse branch sectors, with their capping orientation lines. The signs use the suspended degrees $|x'| = |x| - 1$. Thus a cyclic permutation of the inputs has its usual Koszul sign in these suspended degrees. On a continuous-lift sector the virtual dimension of the underlying disk problem is

$$\text{vdim } \mathcal{P}_{k,\beta} = n + k - 1. \quad (8.5)$$

The capped index gives the corresponding formula on a word containing branch sectors. The operation is zero unless its integrand has top degree. Spaces with $k = 0$ are used for Stokes; the unstable constant disk with just the interior mark is excluded.

Lemma 8.3 (Interior-rooted virtual integration). *Relative to the fixed cyclic boundary-rooted operations, these spaces admit oriented virtual integrations through every finite cutoff. Their normalized boundary products have an interior-rooted parent and an ordinary boundary-rooted child, matched either over the source diagonal or at reciprocal branch sectors. The integrations are cyclic in their external inputs and obey Stokes and composition. In particular,*

$$p_1^\eta(m_0) = 0, \quad p_{1,0}^\eta(c) = \pm \int_L c \wedge i^* \eta \quad (c \in \Omega^2(L)). \quad (8.6)$$

The sign in the second identity is fixed and nonzero. The ordinary disk operations retain their chosen values through the cutoff.

Proof. We construct the interior-rooted spaces with all immersed boundary labels. Their local boundary problems are the ordinary diagonal and transverse-jump problems of [AJ10, Sections 4–5]; the relative construction below specifies their organization when no boundary output is chosen. It uses the mixed parent–child organization of [AFO⁺26, Propositions 16.3 and 16.14], retaining reciprocal internal switches even when the external word is empty. The parent is always additional interior-rooted data. Its identification with a cyclic boundary-rooted operation is neither required nor imposed.

Compactification and normalization. Keep the interior mark in immersed stable-map compactness. At a boundary node the two lifts either agree in L or end at distinct preimages p, q of a double value. In the latter case the two normalized flags have the reciprocal labels (p, q) and (q, p) . These flags disappear from the external word on gluing; they remain part of the normalized stratum. In particular, an empty external word does not restrict its compactification to the diagonal sector.

There are no nonconstant spheres. A connected tree of constant spheres attached at one interior node and carrying at most the one interior mark cannot be stable: it has fewer than three external

special points in total. The degeneration in which the entire boundary collapses would retain a nonconstant sphere, and is also absent. All remaining strata are disk trees. Constant stable vertices are retained, including the disk with the interior mark and one boundary node. The constant disk with no boundary input or node and only the interior mark is unstable and gives no stratum.

Let v_* be the vertex carrying the interior mark. Orient every edge toward v_* . A vertex other than v_* then has exactly one outgoing flag, which is its ordinary disk output; every remaining flag is an input. A normalized corner is the product of the moduli problems on these vertices, with the diagonal matching equations and reciprocal-label matches along its edges. For any collection of edges, cutting them in any order produces this same oriented tree and the same factors. The root of each separated component is either v_* or its unique edge toward v_* . This rule also describes intermediate faces, so the identifications on iterated normalized corners agree.

Stabilization respects these identifications. Forgetting auxiliary marks contracts only constant unstable components. Contracting a non-root such component carries the unique path toward v_* to the unique path toward v_* ; hence the output flag on every surviving ordinary component is unchanged. A constant component carrying the interior mark and an attaching boundary node is stable and cannot be contracted. If it has no such node, it is the excluded constant zero-input root. An unstable constant bivalent disk has either two diagonal nodes on the same sheet or two reciprocal switch nodes: its two constant boundary arcs rule out exactly one switch. Its contraction preserves the labels of the surviving flags. The two-switch constant strip is the identity under the dual cap-line identification fixed by determinant gluing; it is not a new orientation choice [AJ10, Sections 4.3 and 5]. Thus a contraction performs the same pairing of the ordered cap lines whichever order of normalization is used. For a chain of these components, the neck lengths concatenate as $T = T_1 + \cdots + T_j$. This is associative. Forgetting is interpreted in the admissible category of [AFO⁺26, Section 14.3]: its pullbacks of admissible obstruction and CF data are admissible and smooth in the gluing profiles used for integration. The forgetting map itself need not be smooth in ordinary corner coordinates. This admissible compatibility gives compatibility with stabilization as well as with cutting a fixed tree.

These constructions are equivariant under isomorphisms of decorated trees. Such an isomorphism fixes the interior mark and preserves the directed paths to it, so it never changes an ordinary child's output. Relabeling external inputs permutes their positions without rerooting any ordinary child. The same statement holds for an automorphism permuting several isomorphic subtrees. Normalized corners retain their ordering of the cut edges; changing that ordering permutes the outward normal lines with their determinant sign. Their subsequent finite quotients therefore have the required corner identifications and automorphism weights.

Relative charts and smoothing. At a smooth nonconstant root choose auxiliary marked points on compact cores, in free orbits of its finite automorphism group, and impose transverse incidence conditions. Choose obstruction sections supported away from the distinguished mark, the nodes, and the strip ends. Their span can make the augmented Cauchy–Riemann operator surjective: a cokernel vector annihilating all such sections would vanish on an open core and then vanish by unique continuation. Taking the span of the stabilizer translates makes this an invariant finite-dimensional obstruction space. If effective charts are used, include vectors $D_u \xi$ for map variations ξ supported on disjoint free stabilizer-orbit balls in the core. These variations lie in the augmented kernel and carry a faithful stabilizer representation modulo infinitesimal reparametrizations; they make the thickened chart effective. No submersivity is requested at the interior evaluation, which is used only to pull back η .

At a broken root use the direct sum of the obstruction data on its normalized components. Ordinary vertices carry the fixed ordinary data. At a diagonal edge the child output evaluation is submersive, so component surjectivity remains surjectivity after imposing the matching equation in

L. All matching equations can be solved recursively from the root toward the leaves: having chosen the parent variation, choose the child's variation with the required output value and continue with its children. Joint submersivity of all boundary evaluations on one component is not needed. At a reciprocal switch the matching space is a point; the operator has precisely the transverse fixed-jump asymptotic condition of the immersed disk charts. The linear and nonlinear gluing problems are therefore those of the ordinary immersed construction, with the interior-mark coordinate added on the root component. A constant root stabilized by that mark is kept as a marked component in this description.

Transport obstruction sections from the compact cores into the plumbed domains. Exponential gluing estimates give charts through the smoothing parameters, with the corrected smooth gluing profiles cited above. Coordinate changes are obtained from componentwise obstruction data. We specify their relative choice, so that compatibility is more than a choice of unrelated local thickenings. Use the quasi-component construction of [FOOO24a, Conditions 8.12–8.15, Lemmas 8.19–8.20, and Definition 8.21]. A quasi-component records a transported compact-core obstruction choice and its auxiliary stabilization data. On a boundary tree take the componentwise union of the already chosen indices: those of its interior root and the fixed indices of its ordinary vertices. The directed-tree identifications show that this prescription agrees on every iterated boundary and stabilizer quotient.

Extend the corresponding open family of indices inside a proper family over a sufficiently small boundary neighborhood, as in [FOOO24a, Proposition 8.18]. Complete its finite cover by new compact-core choices on the root away from a smaller neighborhood, and saturate the choices under the finite automorphism groups. Choose the new compact-core spaces generically transverse to the finitely many inherited spaces and their stabilizer translates, as in the quasi-component construction. Surjectivity and these direct-sum conditions are then open; shrinking the neighborhoods preserves them. Properness gives the semicontinuity inclusion of obstruction spaces under changes of the central map, as in [FOOO24a, Proposition 8.17]. These inclusions and forgetting the auxiliary marks define the coordinate changes; they satisfy the cocycle conditions because the index inclusions compose and the admissible forgetting operations commute with normalization. The original boundary families are unchanged on the smaller collar. This is the componentwise proof of [FOOO24a, Theorems 5.3 and 5.4], with its additional root data specified relatively. The local diagonal and fixed-jump operators and the augmented matching surjectivity verified above are exactly what that construction uses. The fixed ordinary factors are retained.

Finite extension of perturbations. Let $\delta > 0$ be a lower bound for positive component energies, and choose $0 < \varepsilon < \delta$. Assign cost $E + \varepsilon(k - 1)$ to an ordinary vertex of energy E with k inputs and cost $E + \varepsilon k$ to an interior-rooted vertex. These costs add under boundary splitting. Every ordinary stable vertex has positive cost: the zero-energy unary strip is unstable and represents the separately fixed de Rham differential, and a zero-energy stable ordinary vertex has at least two inputs. The constant one-input interior root has cost ε ; the constant zero-input root is excluded. Each nontrivial breaking therefore has lower-cost factors. The locally finite energy set, the arity cutoff, and stability give a finite collection of problems and normalized corner types, after adjoining all required faces. This includes constant switching polygons; their number of inputs is controlled by arity, not by their zero energy.

Induct on this finite order. On every normalized boundary prescribe the product of the already constructed interior-rooted parent perturbation and the fixed ordinary child perturbation. The prescriptions agree on higher corners by the directed-tree and stabilization identifications. Diagonal fiber products are defined because the child output is strongly submersive for its fixed CF-perturbation; reciprocal switches are products over points. First extend these compatible normalized-boundary products to a corner neighborhood by outer collaring, as in the collar step of [AFO⁺26, Proposition 14.8 and Remark 14.7]. The required agreement on iterated normalized

boundaries is precisely the agreement verified above, including their stabilizer quotients. Apply [AFO⁺26, Proposition 14.5] with the map to a point and with the closed smaller corner neighborhood as its prescribed set. It supplies a compatible thickening and extends the CF-perturbation over the new root space while retaining it near that set. Composition on the prescribed collar still uses the original ordinary child perturbations. There is no requirement to forget a boundary input on the new root space and no new submersion condition at its interior mark.

For cyclic relabeling and finite stabilizers, take invariant obstruction spaces and include the finite-group representations in the auxiliary parameter spaces of the CF-perturbations. The prescribed corner products are equivariant, so the same relative extension is equivariant. This uses equivariant families, not single-valued equivariant transverse sections. Finite quotients are integrated with their ordinary stabilizer weights. Stokes and composition now apply to every prescribed product. The identical construction over a parameter interval, relative to its endpoints, supplies finite-cut comparisons whenever the ordinary choices vary.

Orientations and the zero-input identity. Use the spin determinant orientations and the graded cap lines at all transverse flags. At a reciprocal node the two cap lines are paired in their ordered duality; at a diagonal node they are replaced by the matching determinant of TL . Determinant gluing is associative. Permuting normalized smoothing normals or cap lines gives exactly their Koszul signs, including at an automorphism quotient. Stabilizing marks together with their incidence constraints contribute canceling orientation factors. These are the orientation conventions for ordinary immersed disk composition; adding the even-dimensional interior-mark coordinate does not alter its cap-line rule.

For $k = 0$, pull back the closed $(n-2)$ -form η to the virtual $(n-1)$ -space. Each boundary product has a single ordinary curvature child and an interior-rooted parent with one input. Composition first identifies its contribution with $p_1^\eta(m_0)$, with the common Stokes sign. This includes every reciprocal-switch product. Only after composition do we use the grading: a curvature output has degree two, while each branch generator has odd degree, so those child outputs vanish. There is no sphere or boundary-collapse term. Stokes gives the first identity in (8.6).

At energy zero the one-input diagonal root is the regular constant disk space L . Its one boundary mark and one interior mark fix the domain; its evaluations are the identity and i . Its integral is the second identity in (8.6). The boundary-rooted operations were never changed in this construction. $\square$

8.3 Leading obstructions and canonical models

For a degree-one cochain b of positive valuation, put

$$m_0^b = \sum_{k \geq 0} m_k(b, \dots, b), \quad R_\eta^b(c) = \sum_{r \geq 0} p_{r+1}^\eta(c, b, \dots, b).$$

These sums converge in the energy filtration. The cochain b may have both diagonal and degree-one branch components. Through a fixed cutoff only boundedly many insertions occur.

Lemma 8.4 (Boundary deformations). *For the fixed ordinary operations and the relative integrations of Lemma 8.3,*

$$R_\eta^b(m_0^b) = 0. \tag{8.7}$$

If m_0^b has a first nonzero energy coefficient, that coefficient is a closed exact two-form on L .

Proof. We first establish the identity using the complete virtual boundary products. Apply Stokes to the interior-rooted spaces with k external inputs all equal to b . A boundary child carrying s

consecutive inputs has output $m_s(b, \dots, b)$, and the root carries the remaining $r = k - s$ inputs and this output. Differentiating an input form supplies the zero-energy differential term in m_1 . There is no differential of η and no boundary output differential.

The orientation signs are the suspended insertion signs. At a diagonal matching node this is the ordinary disk Stokes calculation, with the even interior mark in place of the root's boundary-output coordinate. External branch cap lines are moved with their total shifted degrees in the same calculation. Since every input b has suspended degree zero, all insertion signs are $+1$; rotating the degree-two curvature output past these inputs also has sign $+1$. This applies to degree-one point sectors as well as to one-forms. Reciprocal child outputs are retained in the product formula, and then vanish because they would have degree two in an odd-degree point sector.

For $k > 0$ there are k cyclic positions for each contribution. For $0 < s < k$ they are the starts of the child block; for $s = 0$ they are the k gaps; and for $s = k$ they are the k gaps at the child's attaching node, or equivalently its rooted linear orders. The input differentials also have k positions. These are counts on normalized labeled strata, with stabilizer weights retained; we do not quotient the spaces by the external cyclic action. Dividing Stokes by k thus gives

$$\sum_{r+s=k} p_{r+1}^\eta(m_s(b, \dots, b), b, \dots, b) = 0.$$

The separate $k = 0$ identity is $p_1^\eta(m_0) = 0$, proved above. Summing over k proves (8.7). Through any cutoff the sum is finite, so all these operations and divisions are legitimate.

Let c_E be the first nonzero coefficient of m_0^b . Grading places it in $\Omega^2(L)$, since no branch sector has degree two. The leading curved A_∞ equation is $dc_E = 0$. Every term of R_η^b involving a b insertion or positive disk energy has positive valuation. Its remaining term is $c \mapsto \pm \int_L c \wedge i^* \eta$. The coefficient of energy E in (8.7) is therefore

$$\int_L c_E \wedge i^* \eta = 0. \quad (8.8)$$

Choose closed representatives of a basis of $H^{n-2}(D; \mathbb{C})$. Equation (8.8) says that $i_*([c_E] \cap [L]) = 0$. The injection in (8.1) remains injective over $\mathbb{C}$. Poincaré duality on L now gives $[c_E] = 0$. $\square$

Proof of Proposition 8.2. First consider the cyclic canonical model on (8.2). Suppose its curvature has a first nonzero coefficient c_E^{can} . The local finiteness of the energy filtration ensures that a first coefficient exists if the curvature is nonzero. Filtered transfer from this model to the geometric forms model includes an arity-zero component F_0 of positive valuation. Set $b = F_0$. The zero-input functor equation is

$$m_0^{\text{forms}, b} = F_1(m_0^{\text{can}}). \quad (8.9)$$

Its leading term is $F_{1,0}(c_E^{\text{can}})$, where $F_{1,0}$ induces the ordinary cohomology identification. Since the canonical degree-two summand is $H^2(L; \mathbb{C})$, this is a nonzero cohomology class. But Lemma 8.4 says that the first geometric curvature coefficient is exact, a contradiction.

The same argument applies to a canonical structure obtained by promotion from finite cutoffs. Choose a geometric cutoff strictly above E , and compose transfer with the cutoff pseudo-isotopy. Apply Lemma 8.3 relative to the ordinary operations on this geometric representative. Its arity-zero translation still has positive valuation, and its leading unary map still induces an isomorphism on ordinary cohomology. Thus (8.9) through that cutoff gives the same contradiction. All these comparisons use the energy/class filtration of the fixed J_0 disk system. Lemma 8.1 supplies the common gap for the ordinary operations and their comparisons. The interior functionals only need to be constructed on the chosen geometric representative; no identification of them across the

pseudo-isotopy is used. Terms containing positive-valuation cochains require only bounded arity below a fixed energy. Equivalently, one may carry out the argument in filtered $A_{n',k'}$ models, with both cutoffs larger than those needed to detect c_E^{can} . The canonical curvature is therefore zero.

For an embedded test copy there are no positive-energy self-disks at J_0 , so its geometric curvature is zero. Keep J_0 fixed in the pseudo-isotopies and choose transfers with zero arity-zero translation on these objects. The self-disk contributions, including the arity-zero homotopy terms, are empty, so promotion preserves this choice. On a strictly increasing word all morphisms are isolated intersection generators. Energy-zero transfer is the identity on these point complexes, and its restriction needs no operations with repeated labels. The operations on this sector therefore remain the geometric disk operations up to the stated cutoff pseudo-isotopies.

Homotopy units give cohomological units on the resulting uncurved objects. For L this can also be seen by transferring the zero canonical bounding cochain to the forms model: its image is the positive cochain F_0 , which is a bounding cochain by (8.9). Deform the homotopy units there and use the deformed filtered quasi-isomorphism. No strict chain unit is required for this step. All operations are graded, so the cohomological unit has degree zero.

After extending to Λ , the canonical morphism complexes are finite-dimensional. The self complex of L has degree-two chain space exactly $H^2(L; \mathbb{C}) \otimes \Lambda$, because all branch generators have odd degree. Its degree-two cohomology is a subquotient of that space, proving (8.3). Finally, the nondegenerate cyclic pairing gives duality with shift n on morphism cohomology. Its extension to twisted complexes and duality on retracts gives the same pairing on the split-closed triangulated hull, whose Serre functor is consequently $[n]$. $\square$

9 Comparison with the ordinary factor categories

Proposition 8.2 places the immersion and the embedded test branes in a proper uncurved cyclic category. To use the ordinary mirror equivalences, we must identify the category generated by the embedded tests inside that category. The needed comparison concerns only words of distinct, ordered copies. This restriction gives isolated asymptotic intersections at every end, including every internal end of a broken polygon, and removes both self-intersection sectors and unit conventions from the analytic comparison.

Fix one of the finite full cyclic categories constructed in Proposition 8.2, and select its embedded copy labels. Choose any total ordering of these labels and write the copies in that order as

$$K_a = \Sigma_a \times G_a, \quad 1 \leq a \leq N,$$

where each is a copy of a test from Proposition 2.1. Omit Σ_a when Y is a point. Here Σ_a and G_a include the Hamiltonian perturbations and transported brane structures of the copies. As in the preceding construction, all pairs are transverse and there are no triple intersections, also in each factor separately. The fixed split almost complex structure J_0 has no nonconstant spheres, and none of these embedded copies bounds a nonconstant J_0 -holomorphic disk. A *strict word* means a sequence of labels $a_0 < \dots < a_k$. Its inputs lie in the intersection complexes for $(K_{a_{i-1}}, K_{a_i})$, and its output lies in the complex for (K_{a_0}, K_{a_k}) . We initially omit identities and self-morphism spaces. Thus the directed category in this section means precisely these complexes and their operations on strict words.

Proposition 9.1 (Directed comparison). *For each fixed finite full cyclic category above and every total ordering of its embedded copy labels, its strict directed sector is quasi-isomorphic, after an extension of its Novikov field if necessary, to the corresponding ordinary directed Fukaya category of*

the embedded product copies. When Y is present, this ordinary directed category is quasi-isomorphic to the directed restriction of the A_∞ tensor product of the two ordinary factor categories. These comparisons use the specified brane labels and their usual graded intersection complexes. Consequently the ordinary mirror functors in the two factors give a comparison with the directed category on the specified objects $Q_\ell \boxtimes N_M$. All the finitely many ordering comparisons for this same full cyclic category may be made over one field extension.

The two comparisons change different parts of the construction. For the first, we keep J_0 and the intersection generators fixed and change disk integration into regular Floer counts by introducing Hamiltonian terms. Section 9.3 verifies the relative chart and gluing hypotheses for that change. It then removes the energy cutoffs algebraically, while retaining the directed sector of the originally fixed full cyclic category.

For the second comparison, we retain the regular counts and change their domain parameters. The product category uses the same polygon in both factors; the tensor product uses an associahedral diagonal. Sections 9.4–9.6 construct and count a homotopy between these choices, including the boundary and sign identities that make the counts an operad map. Section 9.7 then identifies the result with the tensor product of the ordinary factor categories. All these comparisons concern embedded copies, and ordinary mirror symmetry is used in the individual factors. The freedom to order the same copies in every way will allow us to test compositions through three distinct copies and prove that the copy arrows are invertible in Lemma 10.1.

9.1 Regular Hamiltonian models with fixed intersections

For each ordered pair choose strip data of the form $H(t, x) dt$, $0 \leq t \leq 1$, with H small and identically zero on fixed small balls around the intersections of that pair. Take H sufficiently small on the complement as well. Its chords are then exactly the original intersections. On a stable polygon, with at least two inputs, use a domain-dependent Hamiltonian one-form whose tangential restriction to every Lagrangian boundary arc is zero. The almost complex structure remains J_0 . Product data are the sums of the corresponding data in the two factors.

We require agreement with the pair data on the terminal parts of the external ends and on long necks. On polygon cores the data are smooth up to domain faces and restrict there to the data on the component polygons. More precisely, use the usual consistent strip-like ends and simultaneous plumbing collars on the compactified domain spaces. If T is a neck length, smoothness of the core coefficients is taken in a coordinate $\exp(-cT)$, for a fixed $c > 0$. Thus their difference from the limiting component coefficients, together with every required length derivative, is exponentially small.

These conditions leave enough freedom for regularity. To see their compatibility, first choose data agreeing with their component data on smaller collars and carrying the prescribed strip model near each node. Retain that strip model on terminal end and neck neighborhoods; variations occur on the complementary core patches. The restrictions to intersecting domain faces agree, so extend their coefficients by the ordinary extension theorem for functions on a manifold with corners, using inclusion–exclusion and a partition of unity. The zero tangential boundary condition is linear and is preserved by this extension. At later arities one may shrink the fixed neighborhoods, since the lower-arity data already equal the strip model there. This construction gives smooth extensions with equality on faces; it does not impose equality with component data on an entire domain collar. In particular, at any smooth domain, even one lying in a plumbing collar, there remain open interior core patches on which the Hamiltonian data may vary. The radial plumbing description and strip height used here are also those of [Ope26c, Proposition 7.5].

Lemma 9.2. *For the finite list above, the Hamiltonian data can be chosen small, with the stated face and strip conditions, so that the ordinary strict-word operations are regular. They can also be chosen regular over any prescribed countable collection of smooth parameter chains in products of domain spaces, including all their relative face interiors and all component configurations at those faces. Compactness below an area bound has only polygon and strip degenerations. All nonconstant strips and all contributing stable polygons have positive area, with a common positive lower bound for this finite directed system.*

Proof. The energy identity compares analytic energy with ordinary symplectic area and a curvature term supported on the cores. The integral of that curvature is uniformly bounded for each word, because the pure strip model has zero curvature on every neck. There is no boundary term along a labeled arc, and the chords have not moved. On strips, analytic energy equals area. Areas are additive at broken configurations and are constant in a relative class during the allowed homotopies of data.

At a small-scale bubble the Hamiltonian terms disappear, leaving a J_0 -holomorphic sphere or a disk on one of the embedded copies. Neither exists. Compactness therefore reduces to ordinary Floer compactness with isolated transverse asymptotes. Below a fixed area bound, only finitely many relative classes are realized, even before regularity: near a limiting broken configuration the total relative class is fixed. For regular data this gives finite zero-dimensional counts and compactifications of the one-dimensional moduli spaces.

Nonconstant strips have positive area and a positive energy gap. Stable polygons also have a uniform positive area gap once the data are sufficiently small. Otherwise a sequence of data tending to zero and polygons with areas tending to a nonpositive value would have analytic energies tending to zero. Compactness, including on long necks, would make every component constant. A stable vertex in a strict word has at least three distinct boundary labels, so such a constant component would give a triple intersection. This is excluded. There are only finitely many strict words, which gives a single small-data neighborhood and a common gap.

For strips, regularity follows by varying $H(t, x)$. The regular-point argument based on unique continuation gives dense injective points of $(t, u(s, t))$ on every nonconstant trajectory, away from its asymptotes; it applies with Lagrangian boundary just as in the Floer strip argument of [FHS95], with the corrected linear-algebra step in [FHS24]. The required variations may be supported away from the intersection balls. Indeed a nonconstant strip cannot remain entirely in those balls: the local sheets are exact there, so its area would be zero. A constant strip has an invertible transverse operator.

For polygons, domain-dependent Hamiltonian variations on interior core patches give transversality without a simplicity assumption on the map. The same argument applies over any fixed smooth parameter chain. On a domain face, the vertices in one factor have different tuples of boundary labels, by strictness. Their variations can therefore be chosen independently. A variation on the relative interior of a face extends to the full system by the face-extension construction above; the restrictions on higher families do not constrain variations on these component core patches. Thus the universal linearized problem is surjective on every relevant stratum. Sard–Smale, with countable exhaustions by energy, class, and parameter stratum, supplies simultaneous regularity. Applied to the diagonal chains it gives regular product polygon operations. Product strip regularity is the direct-sum statement for the factor strip operators. $\square$

9.2 Relation to the ordinary Fukaya categories

The preceding data compute the ordinary factor categories used in mirror symmetry. For the graphs this is ordinary weakly exact Floer theory. For the quartic spheres, take their disk-excluding

almost complex structure to be J_0 , as permitted in Seidel's construction [Sei15, Section 8]. It is also sphere-free, so regularity at these fixed data needs no additional bubble-incidence condition. Extend the strict-word data to all ordinary words on the same copies; repeated labels use nondegenerate chord perturbations. The extension is again inductive over domain faces, with the same terminal strip models and smooth core coefficients.

We explain why the slightly more flexible collar condition used above gives the same ordinary category. One can put two choices on separately labeled copies and extend their mixed-word data. Energy and compactness for the comparison follow from the same no-bubbling argument. At a broken curve the core coefficients differ from their limits exponentially in the neck lengths, so ordinary gluing with isolated asymptotes applies. Continuation, binary products, and self-Floer units then identify the two labels of each object. This is the ordinary comparison argument; the collar proof in [Ope26c, Proposition 7.5] uses exactly these hypotheses and even allows decaying errors in strip-end coefficients.

When comparing with an ordinary choice that has varying almost complex data, the quartic factor requires the usual low-dimensional sphere-avoidance argument. The prescribed disk-excluding structure at each boundary label is retained, which excludes boundary bubbles. In real dimension four with $c_1 = 0$, a simple unmarked sphere has expected dimension -2 . Adding the two incidence marks, one on the sphere and one on the principal component, and imposing the four-dimensional matching condition leaves a loss of two dimensions. Removing a sphere tree does not change the principal component index. Hence sphere incidence can be avoided in the zero- and one-dimensional systems used for operations and comparisons. For a multiply covered sphere, its image lies on a simple sphere, so the same incidence exclusion suffices.

The required transversality uses almost complex variations for the sphere and Hamiltonian variations for the principal component, including evaluation at the incidence point. Constant or contracted components do not alter this test. In a relative construction, previously fixed faces already satisfy the exclusion, and the new variations are available on the relative interiors. The terminal strip and neck data may be chosen sphere-free; strip data depending on strip height form only a one-parameter family. Continuation data vary on compact cores, where the same argument applies. These are the sphere-exclusion and comparison mechanisms of [Sei15, Section 8]. Thus the mixed-copy construction identifies our regular systems with conventional ordinary systems, including cohomological units. The actual Hamiltonian sphere copies represent the original objects by [Sei15, Lemma 8.14].

If one uses capped-generator Novikov exponents in the rational-brane normalization, rather than ordinary polygon areas, the coefficients differ by fixed input and output scalar factors. Rescaling the generator lines after the permitted scalar extension makes these descriptions agree. This completes the identification of the regular factor models. It remains to compare them with the disk integration operations that were used to include the immersion.

9.3 The relative comparison from disk integration

Join the zero Hamiltonian data to the regular data by a small path, constant near its endpoints. Keep J_0 and every intersection chord fixed. We use the disk Kuranishi and CF-perturbation construction over this path. Here a CF perturbation is the compatible finite dimensional perturbation used to integrate along a Kuranishi correspondence. The comparison will use its relative extension and Stokes properties; it does not require cyclic perturbations along the path.

Lemma 9.3. *For every finite energy and arity cutoff, the strict-word disk integration operations at the zero end and the ordinary regular counts at the other end are related by a filtered pseudo-isotopy.*

The linear comparison has the identity reduction on the intersection generators. The data at either endpoint may be recovered up to the usual finite-cutoff pseudo-isotopy.

Proof. We verify that the relative disk construction applies to the Hamiltonian terms just introduced. The analytic point is that the unchanged transverse asymptotes and the controlled core coefficients give uniform gluing estimates at every corner. With these estimates, the cited chart and relative perturbation constructions apply with the path parameter included. The inputs are the disk charts from obstruction bundle data, their compatible products on normalized corners, and the relative CF extension and Stokes theorems used in the cyclic construction [AFO⁺26, Sections 9–10 and 12]. For the chart and gluing theorems, we use [FOOO18, Theorem 7.1, Condition 7.11, and Lemma 7.12], the disk-component-wise compatibility construction of [FOOO24a], and the exponential gluing estimates in [FOOO24c, Assumption 3.12 and Theorems 3.13, 6.4, 8.16–8.17]. The following verifications explain their application to this relative problem.

Charts and the path parameter. Use stabilized strip and polygon domains, with the path coordinate included among the parameters. The Hamiltonian terms on a core are smooth lower-order terms in the Cauchy–Riemann equation. Compact-core obstruction sections, transported and summed as in the disk-chart construction, make the linearized equation surjective at a fixed domain and fixed path parameter. Interior test sections suffice: the adjoint unique-continuation argument is unchanged by these smooth lower-order terms. This also makes the path projection submersive. Strip translation is removed by the usual local slice. When stabilization uses auxiliary domain points constrained to target slices, suitable immersive points exist: the maps under consideration have strictly positive area. Strict words give no map isotropy, and the no-bubbling argument requires no sphere chart.

Coefficients near broken configurations. At a gluing seam the ends lie in the balls where the Hamiltonian vanishes. The seam equation therefore coincides with the unperturbed transverse-end equation. A nonconstant strip core down a neck may carry a Hamiltonian term, but that term is translation invariant and transports smoothly with the core.

There is a coordinate issue when a domain neck contains several Floer breaking seams. Its domain length is the total length of the intervening pieces. Consequently its exponential plumbing coordinate is a product of the exponential seam coordinates, times a smooth positive factor. The core coefficients retain their smoothness and exponential derivative estimates after this contraction. Equivalently, in stabilized charts keep the auxiliary marks that locate all the retained cores.

For completeness, use radial variables

$$r_l = e^{-\pi s}, \quad r_r = e^{-\pi(T-s)}$$

on a boundary neck, together with the strip height. Transport to a stabilized strip core multiplies each one-sided radial variable by the intervening radial factors and smooth core factors. Changes of boundary-node coordinate have a positive nonzero leading real coefficient. An origin shift of order $r_l r_r$, after division by one radial variable, still contains the opposite radial factor. The strip coframe thus tends smoothly to the limiting strip coframe in these coordinates. These are the radial collar calculations in [Ope26c, proof of Proposition 7.5]. For coordinate changes between map charts one retains the marks needed for local stability, including marks on strip components. There is no requirement to express the inverse total neck length smoothly in the inverse lengths of a contracted chain.

Uniform gluing estimates. Choose exponential weights below the spectral gaps of the finitely many transverse asymptotic operators. On a joined strip of length T , a smoothed version of $e^{\delta \min(s, T-s)}$ gives a suitable weight. Augment the component operators by kernel observables on compact cores and by the obstruction variables. Pregluing errors are exponentially small in a slightly weaker weight: the maps decay at the ends, the seam equations agree, and the core coefficient errors are exponentially small.

To build a parametrix, split the forcing near each seam midpoint in a fixed-width overlap, solve with the augmented component inverses, and splice over intervals whose lengths are proportional to T . The component one-sided weights dominate the joined weight on their respective sides, uniformly on that overlap. The parametrix error is the sum of an $O(T^{-1})$ cutoff error and an exponentially small coefficient error. There is no matching zero mode at a transverse intersection. A Neumann-series correction therefore gives a uniform augmented inverse; the Fredholm indices agree under gluing. The quadratic estimate is uniform as well, since all new lower-order terms occur on fixed compact cores and the seam equation near the joined values is unchanged.

The differentiated estimates have the same form. Core length derivatives of coefficients are exponentially small. Derivatives of middle cutoffs and reparametrizations multiply exponentially decaying map terms. Passing to weaker weights accommodates every fixed finite number of derivatives, for several necks and with some nodes still broken. Hence the usual disk-chart smoothness in inverse-length corner coordinates applies. These are precisely the exponential estimates required by the cited FOOO gluing theorems. Alternatively, the isolated-end case of [Ope26c, Lemma 4.11 and Proposition 4.12] uses the same augmented surjectivity, spectral gaps, core transports, and exponentially controlled coefficients. Its strip splice estimates are estimates for finite-rank systems and apply in the present dimension. The converse gluing assertion follows from closeness on the cores and decay along the joining ends, giving the local chart coverage.

Compatible charts and relative perturbations. At a normalized corner, choose the product of the component charts over their common path parameter, summing their core-supported obstruction data. At every iterated corner the limiting equation data agree by the face construction. The disk-component-wise compatibility procedure now uses the same stabilizations, nested neighborhoods, transported core sections, openness of augmented surjectivity, and obstruction-data enlargements as for unforced disks. Compactness gives a finite construction at each cutoff. The preceding estimates supply the inverse-function and gluing steps, with the path parameter included. Thus the smooth lower-order core terms introduce no new condition in that construction. There are no forgetful-map or cyclic compatibility requirements on strict words, and all seam matches are at points.

Use the usual outer collars and relative CF extension, preserving the component products and the submersive path projection. Stokes then gives the pseudo-isotopy equations with the required face products. Endpoint chart choices need only be recovered up to a finite-cutoff pseudo-isotopy, as in the ordinary relative construction. At the regular endpoint, a virtual zero-dimensional moduli space has no boundary strata and is already an oriented zero-manifold without isotropy. Its Kuranishi integral is its ordinary signed count, also if one uses a chart with redundant obstruction directions. Thus the endpoint operations are the ordinary operations from Lemma 9.2.

The positive-energy gap makes all nonidentity corrections to the linear comparison have positive valuation. Its reduction is therefore the identity on the unchanged intersection generators. $\square$

All orientations in this comparison come from the same graded spin determinant lines, with zero background class. The comparison of sign conventions is proved in Section 9.6 below. If an ordinary convention reverses arrows, use the opposite category and the corresponding algebraic duality; no reflection of holomorphic disks is involved.

We next remove the cutoffs without imposing simultaneous geometric choices at all orders. Let Λ be the Novikov field in use, with valuation v . There are finitely many strict words in the finite list, and every intersection complex is finite dimensional. Thus the equations for an A_∞ isomorphism between its two directed structures are a finite polynomial system over Λ . Include variables for the inverse of its linear part. For any valuation bound, Lemma 9.3, together with the finite-cutoff comparisons of the chosen promoted cyclic model, provides an approximate solution. Its coordinates have nonnegative valuation, as do the inverse linear-part coordinates, because that linear part has invertible reduction. In particular, the source of these comparisons is the restriction of the fixed full cyclic category. Changing chart or perturbation choices for a particular ordering is accounted for by these endpoint pseudo-isotopies; it does not replace that ambient category.

If this finite system had no solution over an algebraic closure of Λ , the Nullstellensatz would give a fixed polynomial combination of its equations equal to 1. Evaluation at an approximate solution of sufficiently large valuation contradicts this identity: all polynomial coefficients have a fixed lower valuation bound, whereas the equation errors have arbitrarily high valuation. A solution therefore exists after field extension. The argument applies to every ordering of the same finite label set. There are only finitely many orderings, so all their polynomial solutions may be chosen over one algebraically closed extension of the original coefficient field. This proves the first assertion of Proposition 9.1. We now retain the regular ordinary models and compare their product operations.

9.4 Chains of polygon domains

Write $\mathcal{K}_j$ for the compactified space of disk domains with j inputs and one output, $j \geq 2$. It has dimension $j - 2$ and its faces are products of lower polygon spaces. We use its usual smooth face structure from strip-like ends and plumbing. A pair of factor polygons has domain parameter in $\mathcal{K}_j \times \mathcal{K}_j$. The product Fukaya operation uses the diagonal copy of $\mathcal{K}_j$ in this product. Replacing this diagonal by a cellular operad diagonal will give the tensor product.

We need a chain model in which this replacement can be counted geometrically. A geometric chain in $\mathcal{K}_j^2$ is a smooth map

$$c : P \longrightarrow \mathcal{K}_j^2$$

from a compact oriented smooth manifold with corners and finitely many faces. Every relative face interior of P , including its interior, must map into a relative face interior of $\mathcal{K}_j^2$. Impose also the following condition on the way target faces are left. If a face of P has codimension one in a containing face, the inward derivative of c is strictly inward at every target face hypersurface that contains the smaller image interior but does not contain the larger image interior. Derivatives here are taken in the exponential plumbing coordinates. The condition allows several target nodes to be smoothed by one inward parameter.

Take linear combinations of these maps, with the usual orientation, reparametrization, and disjoint-union relations; the boundary is the sum over oriented codimension-one faces. A map that ignores a direct product factor of positive dimension is declared zero. This is a subcomplex relation: if the ignored factor has dimension one, its signed endpoints cancel, and in larger dimensions its boundary factors are still ignored. Products, face inclusions, diagonals, and grafting in both target coordinates preserve the face condition. With the identity in arity one adjoined, these chains form a nonsymmetric differential graded operad, denoted $\mathcal{G}$; a chain of dimension d has cohomological degree $-d$.

Lemma 9.4. *In every arity $j \geq 2$, the chain complex $\mathcal{G}(j)$ has the homology of a point, with the ordinary augmentation in degree zero.*

Proof. Choose a smooth homotopy of $\mathcal{K}_j^2$ that immediately moves every boundary point into the interior, with strictly inward initial derivative at all its incident hypersurfaces, and then contracts within the interior. Such a homotopy is obtained from an inward vector field followed by a contraction of a compact subset of the open cell. Apply it to a chain and use its prism.

At time zero the strict inward condition gives exactly the required condition on the new prism faces. At positive time the image is in the target interior, so no further inward condition remains. The prism identity is therefore valid in $\mathcal{G}$. At the constant endpoint every positive-dimensional chain ignores its whole domain as a product factor and is zero. A zero-dimensional chain is reduced to its augmentation times the chosen interior point. This gives the asserted contraction. $\square$

We can now choose the domain homotopy, before choosing the Floer data with which it will be counted. Let $\mathcal{A}_\infty = C_{\text{cell}}(\mathcal{K}_\bullet)$ be the cellular dg operad of associahedra, with the same cohomological convention. Choose a cellular diagonal

$$\Delta : \mathcal{A}_\infty \longrightarrow \mathcal{A}_\infty \otimes \mathcal{A}_\infty \tag{9.1}$$

lifting multiplication. Such a diagonal exists by induction on arity, or by the free resolution description; see [MS06]. No coassociative choice is required.

There are two maps from $\mathcal{A}_\infty$ to $\mathcal{G}$. The first takes the fundamental chain of $\mathcal{K}_j$ to its actual diagonal in $\mathcal{K}_j^2$. The second applies Δ and then the cross product of geometric fundamental chains. Both preserve the augmentation. Lemma 9.4 and cofibrancy of the $\mathcal{A}_\infty$ operad imply that they are dg-operad homotopic. One can construct the homotopy arity by arity, using polynomial differential forms on an interval as a path object: each obstruction is an augmentation-zero cycle in an acyclic arity.

Fix these two maps and one such homotopy in $\mathcal{G}$. Their coefficients involve only countably many geometric chains. Take the dg suboperad $\mathcal{G}_0 \subset \mathcal{G}$ generated by those chains, and include the representatives and chain relations used in this construction among the choices below. The countable family remains countable after adjoining products, faces, and maps with ignored factors removed. If a relation requires additional representatives, add finite witnesses for a countable basis of the relation space and repeat this closure. This still uses only countably many maps. Choose regular split Hamiltonian data for this family by Lemma 9.2. We require counts only on $\mathcal{G}_0$; no assertion of its acyclicity is needed.

When a parameter lies in a target face, count the polygons at all its vertices, matching the internal ends and summing over their discrete intersection generators. Weight a configuration by its total area. The fixed-domain Fredholm index is

$$|x_0| - \sum_{i=1}^j |x_i|.$$

Consequently a rigid count over a d -dimensional parameter chain acts in degree $-d$.

9.5 Compactification, gluing, and the chain identities

We show that these parametric counts, with the orientation normalization specified in the next subsection, define a dg operad map from $\mathcal{G}_0$ to the endomorphism operations of the directed intersection complexes. This is the geometric assertion needed for the diagonal replacement. It consists of compatibility with the ignored-factor relation, boundary, and grafting.

If a chain ignores a positive-dimensional product factor, a zero-dimensional count on it would have negative expected dimension after that factor is removed. Include the maps with those factors

removed among the regularity requirements. Their moduli spaces are empty, so the ignored-factor relation is respected.

Consider next a one-dimensional parametric moduli space below a fixed area bound. Compactness in Lemma 9.2 permits two phenomena: approach to a face of P and breaking of a Floer strip. Passing to a codimension- r face of P lowers the virtual dimension by r . This remains true when the face creates several nodes in one or both polygon factors. Indeed the fixed-domain indices add under matching at isolated transverse intersections; the domain parameters are supplied by P . Each additional nonconstant quotient strip lowers dimension by one more. Stratum regularity therefore excludes a strip breaking over a parameter boundary in a one-dimensional problem, including a breaking down any newly formed neck. The same argument excludes boundary strata in a zero-dimensional problem. Repeated labels on strip components cause no additional fiber-product transversality requirement, since every match is discrete.

The uniform curvature bounds apply while a chain parameter tends to a face. Thus this is compactness for the parametric problem, not merely compactness at each fixed parameter. Its limiting configurations at a target face are exactly the component polygons and the strip breakings just described.

We must also know that each parameter-face configuration is the end of the one-dimensional moduli space with its prescribed orientation. Use a local collar (t, y) on P , with $t \geq 0$ increasing inward. For every newly smoothed neck its plumbing coordinate has the form

$$\lambda_i(t, y) = t c_i(t, y), \quad c_i(t, y) > 0, \quad (9.2)$$

by the face condition. The neck lengths tend to infinity, and their derivatives in y stay bounded. At $t = 0$, regularity of the nodal problem includes the y variables. Its augmented linear operator is therefore the invertible operator needed for the rigid parametric configuration.

For fixed small $t > 0$, simultaneously preglue all these necks and apply the weighted inverse construction from Lemma 9.3, including the y variables in the linearization. Core coefficients converge to their limiting values. Differentiating in y introduces only bounded length derivatives multiplied by core-supported terms or by terms decaying along the necks. On the component halves, the length-dependent transport has the same exponential estimates. The implicit function theorem gives a unique glued solution for each sufficiently small $t > 0$, and ordinary gluing coverage identifies all solutions near the nodal configuration with this end. Transport of the determinant orientation gives the boundary orientation, with t increasing inward.

This argument uses oriented-end coverage and the inverse estimates for $t > 0$. It does not assert that the solution map is smooth at $t = 0$ in exponential plumbing coordinates. The disk charts used in Lemma 9.3 have the usual inverse-length smoothness; these two uses of gluing require different conclusions.

At an edge already internal in a target face, a broken strip can occur as an output breaking of the child polygon or as an input breaking of the parent. These are the same matched configurations with opposite boundary signs, so they cancel. External strip breakings give the Hom differential. Grafting of chains gives composition by the same discrete matching rule. Finally, on a product of chains in the individual domain spaces, the count factors: regularity makes a rigid product configuration a product of rigid configurations in the two factors. The remaining point is that all these identities use a single consistent sign convention.

9.6 Determinant lines and signs

We use the graded orientation lines of Floer generators and the spin determinant capping and gluing theorem, with zero background class; for the immersed conventions see [AJ10, Sections 4–5]. Our

factor Lagrangian dimensions are even. This parity removes the extra signs that would come from moving the oriented rank contribution of a capped disk.

For a generator x , choose an input cap of index $|x|$ and an output cap of reciprocal index. Their parities agree in even dimension. Let o_x be the input-cap line. Order a fixed map operator first, its input caps in order next, and its output cap last. The spin orientation of the resulting capped disk identifies the map determinant symbol with the graded Hom line

$$\text{Hom}(o_{x_1} \otimes \cdots \otimes o_{x_j}, o_{x_0}).$$

Pair reciprocal cap lines so that the invertible transverse operator on a constant strip gives the identity of o_x .

The determinant gluing comparison at a specified end, with its length held fixed, is local at that end after these caps are inserted. Determinant excision detaches the outer framing and disk contributions; sphere determinants are complex oriented. Only the two node caps remain in the comparison. We record the permutation that recovers the usual Hom sign. Let a child map determinant have parity b , let the sum of the preceding parent input-cap parities be l , and let the node cap have parity q . The sum of the child input-cap parities is $b + q$. Starting with the parent and child determinants, the external input caps, the parent output cap, and the reciprocal node pair, reorder so that each determinant is followed by its own caps. The permutation exponent is

$$bl + q.$$

For example, if r is the sum of trailing parent input-cap parities and o is its output-cap parity, the exponent before simplification is

$$b(l + q + r + o) + q(b + q + r + o) + (r + o)(b + q) = bl + q \quad \text{in } \mathbb{Z}/2.$$

The term depending only on the node, including q , is absorbed by the reciprocal contraction already normalized on the constant strip. Fixed-operator gluing therefore gives Hom composition with Koszul exponent bl . This calculation applies to formal Fredholm capping and gluing, independently of the existence of a rigid holomorphic curve.

For direct sums choose the product cap to be the direct sum of the factor caps, with line $o_x \otimes o_y$. In the order $o_x \otimes o_y \otimes o_x^\vee \otimes o_y^\vee$, its contraction first moves o_y past $o_x^\vee$ and then applies the two factor contractions. The crossing contributes $(-1)^{|x||y|}$, exactly the graded tensor-evaluation sign. The constant product strip is the direct sum of the constant factor strips, so its determinant unit is the tensor of their units and fixes this same reciprocal normalization. At a node the factors $(-1)^{q_x}(-1)^{q_y} = (-1)^{q_x + q_y}$ absorb the product node parity; the intervening mixed permutations remain precisely the tensor Hom signs. Determinant excision and product spin frames give this direct-sum identification, with no additional rank sign in our even dimensions.

For a parameter chain c of dimension d , order its orientation before the map determinant and write $f(c)$ for the resulting raw count. Composing chains of dimensions p and q introduces $(-1)^{pq}$: the second parameter passes the first map determinant. The same sign appears for cross products. Define

$$\gamma(c) = (-1)^{d(d-1)/2} f(c). \tag{9.3}$$

The identity

$$\frac{(-1)^{(p+q)(p+q-1)/2} (-1)^{pq}}{(-1)^{p(p-1)/2} (-1)^{q(q-1)/2}} = 1$$

shows that γ respects both composition and cross products.

For the boundary identity let v be the raw strip map with a fixed translation convention. An output strip places its map determinant before the polygon determinant, so its translation direction passes the d parameter directions. Write $s \in \{1, -1\}$ for its comparison with outward-normal-first orientation before that permutation. At an input the strip translation has the opposite end direction, and its determinant follows the polygon determinant. Moving this translation past both the parameter and the rigid polygon determinant adds no further parity. Thus the oriented boundary equation is

$$f(\partial c) + s(-1)^d v \circ f(c) - s \sum_i f(c) \circ_i v = 0, \quad (9.4)$$

where the slot compositions already include their graded Hom signs. On product generators v has the tensor differential rule.

The same calculation verifies internal-edge cancellation. Put the internal strip determinant in the same position on both descriptions of the broken configuration. Every permutation used to do this is also the permutation in Hom composition, by the fixed-operator calculation above. The two translation directions are opposite, which gives cancellation. At a parameter face, determinant gluing is compared at fixed positive smoothing values. The parameter orientations then compare by their ordinary boundary orientation, with t inward in Equation (9.2). There is no separate normal sign for each node smoothed by that same t .

Set the Floer differential to be sv . Since $f(c)$ has degree $-d$, Equation (9.4) gives

$$d_{\text{End}} f(c) = (-1)^{d+1} f(\partial c).$$

The quotient of the normalization signs in dimensions d and $d - 1$ is $(-1)^{d-1}$. Therefore

$$d_{\text{End}} \gamma(c) = \gamma(\partial c). \quad (9.5)$$

Together with the compactification and grafting arguments, this proves the promised dg operad count identities.

We finally compare this convention with conventional Fukaya signs. Match the cap lines and the order of geometric inputs, using the same dg or suspended translation throughout. The remaining difference is removed by strict sign changes at generators, as in [Ope26c, Lemma 7.7]. Here is the argument in the present setting. Both conventions use the same spin orientation of the capped disk, the same complex orientation on sphere factors, and the same reciprocal contractions. They consequently have no relative-class-dependent ratio. At fixed index the transverse graded-plane data are connected; frame paths with the fixed spin data and determinant winding give the same conclusion. The remaining factors arise from order, marking, and suspension permutations and depend only on the corner indices.

To make the two-vertex argument explicit, use suspended degrees $a_i = |x_i| - 1$ and write the ratio as $(-1)^{r_k(a_1, \dots, a_k)}$, with r_k valued in $\mathbb{Z}/2$. The output suspended degree of a rigid vertex is $1 + \sum_i a_i$. The determinant boundary identity says that

$$r_s(a_i, \dots, a_{i+s-1}) + r_{k-s+1} \left(a_1, \dots, a_{i-1}, 1 + \sum_{j=i}^{i+s-1} a_j, a_{i+s}, \dots, a_k \right) \quad (9.6)$$

is independent of the inserted block and its length. Both conventions have the same suspended insertion sign. Thus Equation (9.6) compares the same oriented one-dimensional formal gluing problem. It applies to arbitrary integer cap indices, without assuming that the associated holomorphic moduli space is nonempty.

Choose $h : \mathbb{Z} \rightarrow \mathbb{Z}/2$ recursively so that $h(a+1) + h(a) = r_1(a)$. Rescaling the line of suspended degree a by $(-1)^{h(a)}$ replaces r_k by

$$r'_k = r_k + h\left(1 + \sum_i a_i\right) + \sum_i h(a_i).$$

It makes $r'_1 = 0$ and preserves Equation (9.6). Comparing a unary child at input i with a unary parent now gives $r'_k(\dots, a_i + 1, \dots) = r'_k(\dots, a_i, \dots)$. Thus r'_k is a constant b_k . Comparing a binary child with a unary parent gives $b_k = b_2 + b_{k-1}$, so $b_k = (k-1)b_2$. A common line rescaling by $(-1)^{b_2}$ removes these constants. This proves a strict sign identification for both the ordinary counts and the directed disk-integration comparison. The integer gradings and even factor dimensions used here are exactly the features needed in this argument.

9.7 The operadic diagonal and the product comparison

Apply the count map γ on $\mathcal{G}_0$ to the homotopy chosen in Section 9.4. At the first endpoint, counts on the actual diagonal are the split product polygon counts. At the second endpoint, factorization on cross products gives precisely the tensor-product operations determined by Δ . For strips the two endpoints have the same tensor differential: a rigid quotient product strip consists of a rigid quotient strip in one factor and a constant strip in the other; if both factors move, the independent translation gives excess dimension.

The construction applies with object labels. The endomorphism operad simply has a component for each composable ordered word, and its compositions use the corresponding slots. Only the bounded arities allowed by the finite strict ordering occur. The operadic path therefore gives isomorphic directed A_∞ structures: integrate its coderivation homotopy successively in arity over the characteristic-zero field. The differential is fixed and there is no arity-one homotopy term, so every step is a finite integration of polynomial coefficients, with identity linear term.

To finish, the cellular-diagonal tensor computes the ordinary tensor through dg models and respects A_∞ quasi-isomorphisms. One may first forget units. Bar–cobar rectification replaces an A_∞ quasi-isomorphism by a zigzag of strict quasi-isomorphisms of operad algebras [LV12]. For fixed-object categories the same construction uses composable tensor words over the finite product of ground fields given by the object idempotents. Units may be adjoined freely when augmented resolutions are used. Tensor these strict diagrams with Equation (9.1); over a field their maps remain quasi-isomorphisms. At the dg endpoints only the associative multiplication survives, and the resulting tensor is the ordinary dg tensor. This also explains the agreement with the tensor-product framework in the product theorem of [Amo17].

Restricting these comparisons to the product labels and strict words preserves every operation and functor equation used here. The ordinary factor mirror equivalences then give the stated directed comparison with their dg mirror categories. This proves the second assertion and completes Proposition 9.1.

10 Completion of the realization construction

We now prove Proposition 2.1. The immersion of Proposition 7.1 has been placed in a proper uncurved cyclic category by Proposition 8.2, and Proposition 9.1 identifies the directed operations on every finite collection of embedded copies with those on the corresponding mirror tests. We first assemble these finite comparisons. Cofinal copies will then recover the full mirror category, to which the immersed object admits an orthogonal projection.

10.1 Copies and a common coefficient field

Retain the ordered Hamiltonian copies chosen in Section 8. Each of the finitely many tests $K_{\ell,M}$ occurs cofinally: after every label there is a later copy of each test. Finite sets in the exhaustion will always include the original tests and at least three copies of each test. Denote the immersed object by L , with its grading and spin data understood.

Lemma 10.1 (Invertibility of copy arrows). *In each finite immersed category containing at least three copies of each test, every nonzero degree-zero cohomology class between distinct copies of the same test is invertible.*

Proof. The strict pair comparison gives a one-dimensional degree-zero cohomology space between copies. Given three distinct copies, choose their directed order in any desired way. Proposition 9.1, ordinary HMS, and Hamiltonian-copy invariance in each factor show that the product of two nonzero composable degree-zero classes between those copies is nonzero.

Let $u : A \rightarrow B$ and $v : B \rightarrow A$ be such classes. Choose a third copy C and a nonzero class $w : A \rightarrow C$. If $vu = 0$, associativity would give $0 = w(vu) = (wv)u$, contrary to the nonzero-product assertion on distinct copies. Thus $vu \neq 0$, and similarly $uv \neq 0$. The self degree-zero cohomology has dimension at most one, because its canonical chain space in that degree is $H^0(K_{\ell,M})$. The cohomological units therefore make vu and uv nonzero multiples of the identities. Rescaling one arrow gives inverse classes. $\square$

The ordinary mirrors of all copies can be fixed once and for all. Indeed, first choose the finite ordinary factor categories on the original tests and their specified images Q_ℓ, N_M . For each larger finite system, compare ordinary choices after including these original labels. Hamiltonian copies are isomorphic to them in the ordinary categories. The finite comparison can consequently use the same mirror objects, with repeated labels. No analytic functor on an infinite collection is needed.

The finite cyclic categories and their comparisons need not be compatible as the label set increases. We assemble them by a field ultraproduct: the canonical complexes bound each Hom dimension, and every equation involving fixed labels is a finite matrix identity. These two facts allow all the required structures to pass to one coefficient field in the next lemma.

Lemma 10.2 (Simultaneous finite comparisons). *Over an extension field $\mathbb{k}$ of characteristic zero and cardinality at most the continuum, there is a proper uncurved cyclic A_∞ category containing L and all the test copies. Its directed subcategory on the test copies admits an A_∞ functor to fixed mirror models, inducing isomorphisms on the cohomology of every strict arrow. The degree-two bound for L is*

$$\dim_{\mathbb{k}} \text{Hom}^2(L, L) \leq b_2(D) - \dim Z. \quad (10.1)$$

The category has cohomological units and a nondegenerate cyclic pairing of dimension n .

Proof. Exhaust the labels by finite systems. For each one, take the uncurved A_∞ category of Proposition 8.2, with its bound (8.3), and the directed comparison of Proposition 9.1, extending its coefficient field when necessary. Pass to minimal models. The immersed models can be chosen cyclic: a finite complex with a nondegenerate compatible pairing admits a splitting and a contraction compatible with that pairing. Transfer the directed functors to these models and to the fixed mirror models on the repeated labels.

For any fixed pair of labels, the dimensions and degrees of the minimal Hom spaces are bounded by the ranks of the corresponding finite canonical complexes. Choose a nonprincipal ultrafilter on the exhaustion. For each fixed Hom, its graded ranks are constant on a set in the ultrafilter. Use bases of those ranks and take the ordinary field ultraproduct of the coefficient fields. Every fixed A_∞

equation and functor equation is a finite matrix identity, so passes to this field. The same applies to units, cyclicity, nondegeneracy of the pairings, invertibility of the strict-arrow cohomology maps and of the copy arrows, and (10.1). The fixed mirror models undergo ordinary scalar extension along the diagonal embedding of their common ground field.

The finite canonical ranks show that the resulting category is proper. All fields in the finite constructions can be taken to have cardinality at most the continuum, and a countable ultraproduct has the same bound. There is therefore an abstract embedding of the resulting field into $\mathbb{C}$. The construction has used ordinary finite-object HMS before taking this ultraproduct; it makes no assertion of analytic mirror symmetry over an ultraproduct field. $\square$

10.2 Recovering full morphisms from directed copies

We explain why the directed comparison determines the enhanced full subcategory of copies. The two receiving categories are the full subcategory of test copies in the immersed category and the full subcategory of their specified mirror images, with repeated labels. Replace both receiving categories by strictly unital models, transport the functors on strict words, and adjoin formal units to the directed category. Write $\mathcal{D}$ for this unital directed category, and let $\mathcal{C}$ denote either full category of copies receiving it. The receiving functor is a quasi-isomorphism on strict arrows, and copy arrows become isomorphisms in cohomology by Lemma 10.1 and its ordinary mirror counterpart.

For a label j , write $h_j^{\mathcal{D}} = \text{Hom}_{\mathcal{D}}(-, j)$ for its right Yoneda module. Choose successive later copies $j_0, j_1, \dots$ of j and degree-zero copy arrows between them. Restriction of the full representable admits a natural comparison

$$\text{hocolim}_k h_{j_k}^{\mathcal{D}} \longrightarrow \text{Hom}_{\mathcal{C}}(-, j)|_{\mathcal{D}}. \quad (10.2)$$

To construct it, first take the same telescope of restricted full representables. All its arrows are equivalences, so its homotopy colimit is the right-hand side. The Yoneda comparisons from directed representables to restricted full ones commute with the arrows up to the natural homotopies and induce (10.2). At a fixed argument label i , every sufficiently late j_k satisfies $i < j_k$, where the comparison on Hom is a quasi-isomorphism. Thus (10.2) is a quasi-isomorphism on every argument.

Taking Hom from this telescope to another restricted full representable gives the homotopy limit of its values at j_k . The transition maps in that homotopy limit are equivalences, because the full-category copy arrows are invertible. It follows that restriction of Yoneda is fully faithful on the full copy category. This argument applies to higher A_{∞} functors as well, using their natural Yoneda comparisons.

On both sides the restricted representable of each original test is therefore the same telescope in the derived module category of $\mathcal{D}$. The enhanced split-closed triangulated hulls of the two full categories consequently agree. By Lemma 2.4 and sphere generation, their mirror hull is

$$\mathcal{A} \simeq \text{Perf}(B \times E^g). \quad (10.3)$$

Here we use ordinary external-product generation for smooth projective varieties: external products of generators of the two factors generate the perfect category of their product.

10.3 Projection and the Euler pairings

Let $\mathcal{F}$ be the split-closed triangulated hull of the immersed category including L . Its nondegenerate cyclic pairing makes it proper with Serre functor $[n]$; the pairing extends to twisted complexes and then to retracts. The full subcategory $\mathcal{A}$ in (10.3) is saturated [BV03]. The module $\text{Hom}_{\mathcal{F}}(-, L)|_{\mathcal{A}}$

has finite-dimensional total cohomology at every object, by properness. Saturation therefore represents it by an object $\mathcal{E} \in \mathcal{A}$. The representing map fits into a triangle

$$\mathcal{E} \longrightarrow L \longrightarrow R \longrightarrow \mathcal{E}[1], \quad \mathrm{Hom}^*(\mathcal{A}, R) = 0.$$

Serre duality in $\mathcal{F}$ gives $\mathrm{Hom}^*(R, \mathcal{A}) = 0$ as well. The connecting map vanishes, and the triangle splits. In particular $\mathcal{E}$ is a summand of L , so

$$\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \leq \dim \mathrm{Hom}_{\mathcal{F}}^2(L, L) \leq b_2(D) - \dim \mathcal{Z}.$$

After the abstract field embedding, this proves (2.5) on the complex mirror.

For every test object, the orthogonal summand R has zero Euler pairing. The Euler pairing with L is computed by its transverse Floer complex, hence equals the oriented intersection pairing with the immersed fundamental class, up to the common sign fixed by the grading convention. Equation (7.1) says $\mathrm{PD} \, i_*[L] = \nu a$ with $\nu > 0$. It follows that all the test Euler pairings equal the pairings with a times one common nonzero rational scalar. This is (2.6) and completes the proof of Proposition 2.1.

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